



**Verified Carbon
Standard**

SINOP LANDFILL GAS TO ELECTRICITY PROJECT



Document Prepared by:

LGAI Technological Center S.A. (Applus+ Certification)

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Summary:

LGAI Technological Center S.A. (Applus+ Certification) (Hereafter referred as Applus+ Certification) has been appointed by “Biokütle Enerji Elektrik Üretim Anonim Şirketi” to perform the Joint validation and verification of the “Sinop Landfill Gas to Electricity Project” is being implemented by “Biokütle Enerji Elektrik Üretim Anonim Şirketi” (VCS ID 4014¹). Project located at Sinop Landfill Site in Hacıoglu Neighborhood, Central District, Sinop Province Turkey.

The project activity is primarily aimed at generating electricity through landfill gas-based engines and subsequently making a contribution towards mitigating greenhouse gas (GHG) emissions into the atmosphere through avoidance of same power generation by fossil fuel power plants. The project activity is a significant step towards meeting the energy demands of the country which is facing an acute shortage of power through use of renewable energy sources. The power generated is exported to the national grid. The project has feeder at substation. Also, project activity helps to reduce GHG emission by avoiding decomposition of MSW (Municipal Solid Waste) in a solid disposal site that would have occurred in the absence of the project activity and anaerobic decomposition of organic matter through landfill gas to electricity generation process.

Validation Summary: The validation objective is an independent assessment by a Third Party of a proposed project activity against all defined criteria set for the registration under the VCS. In order to confirm that the project activity, as documented, is sound reasonable and meets the identified criteria, the validation involves the assessment of project conformance to VCS standards/programs, project conformance to the applied methodology, including the procedure for the demonstration of additionality specified in the methodology; and likelihood that methods and procedures set out in the Joint Project Description and Monitoring Report (hereafter refer to as “joint PD & MR”) will generate verifiable GHG data and information when implemented. Validation is a requirement and is seen as necessary to provide assurance to stakeholders of the quality of project and its intended generation of VCU. Validation is part of the VCS project cycle and will finally result in a conclusion by the executing VVB whether a project activity is valid to be submitted for registration to Verra registry. Validation is conducted using LGAI Technological Center S.A. (Applus+ Certification) procedures in line with the requirements specified in the latest version of the VCS Validation and Verification Manual and applying auditing techniques. The assessment team assessed the project activity’s compliance against the VCS Standard Version 4.4, the selected CDM methodology and the joint PD & MR. The project is eligible under non-AFOLU project.

Validation purpose: The project is registered with VCS Project ID 4014. The start date of the project activity is the 01-July-2021. This is the earliest date of commissioning of the project activity. The First Crediting Period was from 01-July-2021 to 30-June-2028. The project activity is estimated reducing GHGs to the 40,803 tCO₂e/annum for the entire crediting period of 07 years which is twice renewable. (Total 21 years.)

Verification Summary: LGAI Technological Center S.A. (Applus+ Certification) has performed the verification of the emission reductions reported for the project activity “Sinop Landfill Gas to Electricity Project” (VCS Project ID: 4014) for the monitoring period 01-July-2021 to 28-February-2023 to review and determine the monitored reductions in GHG emissions that have occurred as a result of the project activity. These emission reductions are claimed as Verified Carbon Units (VCU) under the Verified Carbon Standard (VCS) version 4.4. The verification was performed on the basis of VCS Program Guide version 4.3 and VCS standard version 4.4 for the VCS projects, as well as criteria given to provide for

consistent project operations, monitoring and reporting. The verification was conducted by means of document review, follow-up interviews and site inspections, and the resolution of outstanding issues.

Verification purpose: The main purpose of this verification activity is to have an independent third party for the assessment of the project design, monitoring report to ensure a thorough assessment of the proposed project activity against the applicable CDM and VCS requirements.

The monitoring period covered under this periodic verification is 01-July-2021 to 28-February-2023 (including both days) and the project activity achieved a total 41,132 tCO₂e emission reductions during this monitoring period from the generation-mix of power plants connected to the Grid, which is mainly dominated by thermal/fossil fuel-based power plant.

The scope of the verification is the independent and objective review of the Joint Project Description (PD) and Monitoring Report (MR). The joint PD & MR is reviewed against the relevant criteria (see above) and decisions by the CDM Executive Board and VCS executive board, including the approved baseline and monitoring methodology/3/. The verification was based on the guidance given in the CDM validation and verification standard for project activities, version 03.0, review against previous registered project documents, VCS program guideline version 4.3 and VCS Standard version 4.4.

A risk-based approach has been followed to perform this validation and verification activity. In the course of verification, 02 Corrective Action request (CAR) and 07 Clarification Requests (CLs) were raised and successfully closed. No FAR was raised during this validation and verification. The review of the joint PD & MR and additional documents related to baseline and monitoring methodology /3/; the subsequent background investigation, follow-up interviews and PP have provided LGAI Technological Center S.A. (Applus+ Certification) with sufficient evidence to verify the fulfilment of the stated criteria of VCS.

The assessment team has employed a risk-based approach to assess the completeness and accuracy of the claims and conservativeness of the assumptions in the MR. The main focus of the assessment team is to identify the significant risks for the project implementation and the generation of VCUs. The validation and verification are not meant to provide any consulting towards the project participants. However, stated requests for clarifications and/or corrective actions may have provided input for improvement of the PD-MR combined.

The only purpose of the verification is its usage during the issuance process as part of the VCS project cycle. Therefore, LGAI Technological Center S.A. (Applus+ Certification) can't be held liable by any party for decisions made or not made based on the verification opinion, which will go beyond that purpose.

The validation and verification have been planned and organized to achieve a Reasonable Level of assurance as per the requirement of VCS. No sampling procedure applied for document verifications. The entire documents checked/verification conducted to arrive at positive verification conclusions.

CONTENTS

1	INTRODUCTION	6
1.1	Objective	6
1.2	Scope and Criteria	6
1.3	Reasonableness of Assumptions and Level of Assurance	7
1.4	Summary Description of the Project	7
2	VALIDATION AND VERIFICATION PROCESS	8
2.1	Method and Criteria	8
2.2	Document Review	10
2.3	Interviews	10
2.4	Site Visits	11
2.5	Resolution of Findings	11
3	VALIDATION FINDINGS	12
3.1	Project Details	12
3.2	Participation under Other GHG Programs	19
3.3	Safeguards	20
3.4	Application of Methodology	23
3.5	Non-Permanence Risk Analysis	54
4	VERIFICATION FINDINGS	55
4.1	Accuracy of GHG Emission Reduction and Removal Calculations	55
4.2	Quality of Evidence to Determine GHG Emission Reductions and Removals	61
5	VALIDATION AND VERIFICATION OPINION	63
	APPENDIX 1: DOCUMENTS REVIEWED OR REFERENCED (VERIFICATION)	67
	APPENDIX 2: CORRECTIVE ACTION REQUESTS, CLARIFICATION REQUESTS AND FORWARD ACTION REQUESTS (CAR/CL/FAR)	69
	APPENDIX 3: COMPETENCE OF TEAM MEMBERS AND TECHNICAL REVIEWERS	75
	APPENDIX 4: ABBREVIATIONS	77

1 INTRODUCTION

1.1 Objective

LGAI Technological Center S.A. (Hereafter referred as Applus+ Certification) has been appointed by “Biokütle Enerji Elektrik Üretim A.Ş.” to perform the validation and verification of the “Sinop Landfill Gas to Electricity Project.” under guideline version 4.3 and VCS Standard version 4.4. The objective of this validation and verification activity is to have an independent third party for the assessment of the project design, Joint PD & Monitoring Report and Final Joint Validation and Verification report and to ensure a thorough assessment of the proposed project activity against the applicable CDM and VCS requirements. In particular.

- The project's baseline is assessed against AMS-III.G Version 10 /3/. and AMS-I.D Version 18.0 /3/
- The projects compliance with the requirements of Article 12 of the Kyoto Protocol, the CDM Modalities and Procedures as agreed in the Marrakech Accords under decision 3/CMP.1, the annexes to this decision, subsequent decisions and guidance made by COP/MOP & CDM Executive Board and other relevant rules, including the Host Country legislation and sustainability criteria along with VCS guideline version 4.3 and standard version 4.4.
- CDM validation and verification standard for project activities, Version 03.0
- VCS program guideline v 4.3
- VCS standard v.4.4

Joint validation and verification are seen as necessary to provide assurance to stakeholders on the quality of the project and its intended generation of verified Carbon Units (VCUs) for the renewed crediting period & verification is a requirement for all VCS projects and is seen as necessary to provide assurance to stakeholders of the quality of the project and its intended generation of Verified Carbon Units (VCUs).

1.2 Scope and Criteria

The scope is defined as an independent and objective review of the Joint VCS PD and MR is prepared as per the approved methodology AMS-III.G Small-scale Methodology: Landfill methane recovery. Version 10.0 and AMS-I.D Grid connected renewable electricity generation. Version 18.0. The Joint VCS PD and MR is reviewed against the criteria stated in Article 12 of the Kyoto Protocol, the CDM modalities and procedures as agreed in the Marrakech Accords and the relevant decisions by the CDM Executive Board and VCS standard Version v4.4. and guideline version 4.3 including the approved baseline and monitoring methodology AMS-III.G Version 10.0 and AMS-I.D Version 18.0. The validation & verification was based on the requirements in the CDM validation and verification standard for project activities, Version 03.0. and VCS program guideline version 4.3. and VCS Standard version 4.4. The validation & verification is not meant to provide any consulting towards the project participants. However, stated requests for

clarifications and/or corrective actions may have provided input for improvement of the Joint VCD PD and MR. In line with Guidelines for Application of materiality in validation & verifications, the assessment team has conducted a complete verification of all the information presented in the Joint VCS PD and monitoring report and data presented ex-ante Emission reduction sheet and monitored data presented in the emission reduction calculation spread sheet. It invoices follows the paper trail back to the raw data such as meter reading records and invoices. There are no material errors, overestimation of ER, omission or misstatement. The validation/verification team has reviewed all the documents like commissioning certificates etc.

1.3 Reasonableness of Assumptions and Level of Assurance

The joint validation and verification report is based on the joint PD & MR, supporting evidences made available to the VVB and information collected through performing interviews and during the On-site visit (OSV).

The validation conclusion is based on reasonableness of the assumptions, limitations, and methods that support the statement about the outcome potentially achieved by the activities during operation and monitoring, provided that they may change during the mentioned stages of the project's development and operations.

The verification has been done with a reasonable level of assurance about whether the reported GHG emissions reduction data is free from material misstatement.

The joint validation and verification report expresses a conclusion with a reasonable level of assurance about whether the reported GHG emissions reduction data is free from material misstatement. LGAI Technological Center S.A. (Hereafter referred as Applus+ Certification) applied a materiality threshold of 5% with respect to omission or misstatements concerning reported quantities as per VCS standard.

1.4 Summary Description of the Project

Sinop Landfill Gas to Electricity Project, where landfill gas from the landfill of Sinop used to generate renewable power. The Landfill Gas to Electricity power project has been developed using existing landfill site for the project activity.

Electricity production is done by burning the landfill gas. Gas engine with an installed capacity of 2.906 MWm / 2.828 MWe utilizes collected landfill gas for power generation. The project began operation in 01-July-2021. It will handle an average of 173.6² tons MSW per day (average) and further expected to rise during operation period from 155 t/day in 2021 and 236 t/day in 2040. Waste consists of organic materials such as pulp, paper and cardboard, and food waste. The start date of the project activity is the 01-July-2021. This is the earliest date of commissioning of Gas engine. The expected annual emission reductions during the crediting period are 40,803 tCO₂e. In current monitoring period from 01-July-2021 to 28-February-2023 (First and last date included) the project activity has resulting in GHG emission reductions of 41,132 tCO₂e.

² Feasibility Report of Sinop Landfill Gas to Electricity Generation Facility

2 VALIDATION AND VERIFICATION PROCESS

2.1 Method and Criteria

Validation and Verification Scope: The scope is defined as an independent and objective review of the Joint PD & Monitoring report. The VCS PD and MR is reviewed against the criteria stated in Article 12 of the Kyoto Protocol, the CDM modalities and procedures as agreed in the Marrakech Accords and the relevant decisions by the CDM Executive Board and VCS standard, Version 4.4. and guideline version 4.3., including the approved baseline and monitoring methodology AMS-III.G Version 10. and AMS-I.D Grid connected renewable electricity generation --- Version 18.0. The validation and verification were based on the requirements in the Validation and Verification Standard for project activities, version 03.0., and VCS guideline version 4.3. and standard, version 4.4.

The validation and verification are not meant to provide any consulting towards the project participants. However, stated requests for clarifications and/or corrective actions may have provided input for improvement of the combined project description and the Monitoring report.

Validation and Verification Process: The project assessment is based on the Clean Development Mechanism Validation and Verification Standard for project activities, version 03.0. and VCS Standard version 4.4. and VCS guideline, version 4.3. and is conducted using standard auditing techniques to assess the correctness of the information provided by the project participants. Before the assessment begins, members of the team covering the technical scope(s), sectoral scope(s), and relevant host country experience for evaluating the VCS project activity are appointed.

Once the project is received by the assessment team, the members of the assessment team carried out:

- I. A desk review of the Joint PD and MR;
- II. Onsite Visit with project participant;
- III. The resolution of outstanding issues and the issuance of the final verification report and opinion.

The prepared Joint validation and verification report and other supporting documents then undergo an internal quality control before being submitted to the VCS executive board.

In order to ensure transparency, assumptions must be clear and stated explicitly and background material must also be referenced. LGAI Technological Center, S.A. (Applus+ Certification) has developed a specific checklist customized for the project. The checklist demonstrates, in a transparent manner, the project criteria (requirements), discussion on each criterion by the assessment team, and the results from verifying the identified criteria.

Appointment of the assessment team

According to the sectoral scope / technical area and experience in the sectoral or national business environment, LGAI Technological Center S.A. (Applus+ Certification) has composed a project assessment team in accordance with the appointment rules in the internal Quality Management System of LGAI Technological Center S.A. (Applus+ Certification).

The composition of audit team shall be approved by the LGAI Technological Center S.A. (Applus+ Certification) ensuring that the required skills are covered by the team.

The four qualification levels for team members that are assigned by formal appointment rules are as presented below:

- Lead Auditor (LA).
- Auditor (A) / Auditor in Training (AiT).
- Technical Expert (TE).
- Technical Reviewer (TR).

The sectoral scope / technical area knowledge linked to the applied methodologies shall be covered by the assessment team.

Name	Role	SS Coverage	TA Coverage	Financial aspect	Host country experience
Dr. Atul Takarkhede	Lead Auditor/Technical Expert	YES	YES	NA	No
Mr. Jitendra Mohan Singh	Country Expert	YES	YES	NA	YES
Mr. Denny Xue	Technical Reviewer	YES	YES	NA	NA

The complete list of CVs is included as Appendix 3 of this report.

Document review

The Joint PD and Monitoring Report Version 07 dated on 04-June-2024 submitted by the PP was reviewed against the approved methodology and other relevant criteria to verify the correctness, credibility, and interpretation of the presented information. Furthermore, a cross-check between information provided and information from other sources has been done. A complete list of all documents and evidence material reviewed is included in this report below in Appendix 1.

Onsite Visit

An onsite audit was conducted by LGAI Technological Center S.A. (Applus+ Certification) who performed site visit on 14-May-2023 and 15-May-2023 with project stakeholders to confirm selected information

and to resolve issues identified in the document review. The detail is provided in this report in the sections 2.3 and 2.4 below.

Resolution of Clarification and Corrective Action Request

The objective of this phase of the Verification was to resolve the requests for corrective actions and clarification and any other outstanding issues which need to be clarified for Applus+ Certification positive conclusion on the Monitoring report. The Corrective Action Requests and Clarification Requests raised by Applus+ Certification were resolved during communications between the Client and Applus+ Certification to guarantee the transparency of the verification process, the concerns raised and responses given are summarized below in the Appendix 2.

The final Joint PD and MR Version 07 dated 04-June-2024 submitted by PP serves as the basis for the final assessment presented. Additional changes to the project during the verification process are not considered to be significant with respect to the main CDM/VCS objectives. The two CDM/VCS main objectives are the reduction of anthropogenic GHG emissions and the contribution of sustainable development to the host country.

Internal quality control

As final step of a Joint Validation & Verification of the final documentation including the final Joint validation & verification report and the checklist have to undergo an internal quality control by the technical review committee, i.e., each report has to be finally approved either by the head of the technical review committee or the deputy. In case one of these two persons is part of the assessment team approval can only be given by the other one to avoid any conflict of Interest.

After confirmation of the project owners the positive verification opinion and relevant documents are submitted to the VCS secretariat through the VCS web-platform.

2.2 Document Review

The joint PD & MR and supporting background documents related to the project implementation were reviewed. Documents review was conducted to ensure consistency with and identify any deviation from VCS program requirements. Desk review included an examination of the project design details, baseline scenario, additionality, ex ante and monitoring data and parameters, and ex ante quantification of GHG emission reductions. The assessment team also used additional documentation by third parties like host party legislation, technical reports referring to the project design details, baseline scenario, additionality, monitoring or to the basic conditions and technical data. Furthermore, the joint PD & MR and emission reduction calculation spreadsheet of monitoring period from 01-July-2021 to 28-February-2023, relevant decisions, clarifications and guidance from the Verra, other information and references relevant to the project activity have also been assessed during the desk review.

The details of the document observed during the validation and verification process are listed in Appendix 1 of this report.

2.3 Interviews

Interviewee		Date	Subject	Team member
Name	Affiliation			
Mr. Mushin Dugan	Site Manager	14-May-2023 and 15-May-2023	Project Implementation, procedure, TEAIS reports, calibration, grievance mechanism, Management practices, data storage, QA/QC	Dr. Atul Takarkhede (Lead Auditor) Mr. Jitendra Mohan Singh (Country Expert)
Mr. Doğukan Arıkan	Expert.		GHG calculations, MR and ER preparation, Data collection, data storage, QA/QC	
Mr. Ahmet Darkan	Head of the Village		Local Stakeholder Consultation	
Mr. Senol Ince	Local Villager			

2.4 Site Visits

Duration of On-site inspection: 14-May-2023 and 15-May-2023				
No.	Activity performed on-site	Site location	Date	Team member
1.	Assessment team checked the implementation of the project, Baseline emission, Emission reduction calculation, technical description of the project and Monitoring. Assessment team also checked that whether the monitoring plan as described in the VCS PD and MR is actually practised onsite. Also, assessment team checked any change in host country criteria which may affect the baseline of the project activity.	Sinop Landfill Site in Hacıoglu Neighborhood, Central District, Sinop Province Turkey.	14-May-2023 and 15-May-2023	Dr. Atul Takarkhede (Lead Auditor) Mr. Jitendra Mohan Singh (Country Expert)

The interviewed personnel during on site visit and objective are listed in above table.

2.5 Resolution of Findings

The objective of this phase of the Validation and Verification was to resolve the requests for corrective actions and clarification and any other outstanding issues from previous validation which need to be clarified for Applus+ Certification's positive conclusion on the Monitoring report. The Corrective Action Requests and Clarification Requests raised by Applus+ Certification were resolved during communications between the Client and Applus+ Certification to guarantee the transparency of the verification process, the concerns raised, and responses given are summarized below in the Appendix 2.

The final joint PD & MR Version 05 submitted by PP serves as the basis for the final assessment presented. Additional changes to the project during the verification process are not considered to be significant with respect to the main CDM/VCS objectives. The two CDM/VCS main objectives are the reduction of anthropogenic GHG emissions and the contribution of sustainable development to the host country.

Areas of validation and verification findings	No. of CL	No. of CAR	No. of FAR
Project design document and Monitoring report	CL01,	00	00
Description of project activity	CL02, CL03, CL05, CL07	CAR01	00
Application of selected baseline and monitoring methodology and selected standardized baseline			
Applicability of methodology and standardized baseline	00	00	00
Deviation from methodology	00	00	00
Clarification on applicability of methodology, Tool and/or standardized baseline	00	00	00
Project boundary	00	00	00
Establishment and description of baseline scenario	00	00	00
Demonstration of additionality	00	00	00
Emission reductions	00	CAR02	00
Calibration details	00	00	00
Monitoring plan	CL04	00	00
No Net harm assessment	00	00	00
Local stakeholder consultation	CL06	00	00
Others (please specify)	00	00	00
Total	07	02	00

The list of findings and the resolution is presented in Appendix 2 of this report.

2.5.1 Forward Action Requests

This is initial validation + verification process, hence No FAR was raised previously.

3 VALIDATION FINDINGS

3.1 Project Details

The project activity is Sinop Landfill Gas to Electricity Power Plant. It is a landfill gas to electricity power project which is utilizing landfill gas from landfill site to generate renewable power.

The expected average annual emission reductions are 40,803 tCO₂e/y by project activity. The generated electricity will be supplied to grid. The project activity help in greenhouse gas (GHG) emission reduction by using Landfill Gas for generating electricity which otherwise would have been generated using grid mix power plants, which is dominated by fossil fuel based thermal power plants.

The purpose of the project is to utilize landfill gas (LFG), which consists mainly of methane, for electricity generation. It includes LFG collection system, LFG pre-treatment system, electricity generation system and gas collection system.

The landfill gas to electricity project consists of:

- 1- Pipelines that collect landfill gas from the landfill with the help of the Booster.
- 2- Booster system that pulls LFG
- 3- Blower that blows the LFG to the Gas Engines
- 4- Gas engines that burn the LFG to produce electricity
- 5- Gas ballon system (3500 m³) that collects the extra gas that is not burnt in the Engines.

The working principles of landfill gas are as follows. The LFG caused by the organic content of the landfill is collected by pipelines connected to the booster system. The gas which Booster collects also blew the gas to the engine to be burnt down. If there is a failure of the engines, the collected gas is directed to the Gas Ballon collection system. In the system, there is no need for auxiliary fuels for the start-up of the engines. The LFG can be burnt directly.

LFG collection system

LFG is extracted from MSW under negative pressure by blower pumps and moved through wells, then collected by gas collection stations and transferred by sub-pipes and a main pipe to LFG treatment equipment. Flow rate of the LFG is regulated at the collection points in order to always fit with the consumption capacity of the generation engines.

LFG treatment system

Prior to electricity generation, LFG is treated to remove impurities and moisture, to avoid corrosion in the engines. The treatment consists of filtration, de-moisturing, cooling and pressurization.

Electricity generation and grid connection system

2 generators are fed with the LFG to generate electricity, however only one motor engine and generator is in operation other one will be in operation by 2024, which is then exported to the grid. Before the implementation of the project activity, no gas extraction and utilization systems existed and the LFG was emitted into the atmosphere directly. The project collects and utilizes LFG to generate electricity via gas engines and supply to Turkish' state utility.

The project activity is located in within Sinop Solid Waste Landfill Facility Sinop, in Hacıoglu Neighborhood, Central District, Sinop Province Turkey. During Site visit, Geo coordinates checked on Google earth and verification team conclude geo coordinates of project locations are consistent with Joint PD and MR.

The details of Project's commissioning and location are as follows:

Name of the Project Promoter	Location	Geographical Co- ordinates		Commissioning Date
		Latitude	Longitude	

Sinop Landfill Gas to Electricity Project	Sinop Landfill Site in Hacıoglu Neighborhood, Central District, Sinop Province Turkey.	41.949269823	35.036668530	29-June 2021
		41.949272309	35.036911608	
		41.949194300	35.036913527	
		41.949194861	35.036979850	
		41.948934092	35.036983778	
		41.948930761	35.036673646	

Technical specifications:

Technical Specifications - Gas Engine (which already installed)	
Manufacturer	Jenbacher Innio
Model	J420GS
Motor Power	1453 kW
Rotation	1500 min ⁻¹
Year of manufacture	2021
Life time	25 years
Technical Specifications – Gas engine (which will be install in 2024)	
Manufacturer	Jenbacher Innio
Model	J420GS
Motor Power	1453 kW
Rotation	1500 min ⁻¹
Year of manufacture	2021
Life time	25 years

Gas Balon technical specifications

Manufacturer/Supplier	Ecomembrane
Model	2MASTER®
External Shape	¾ sphere
Production year	2021
Volume/gas holding capacity	3500 m ³

The project activity doesn't involve any GHG emission sources. This project adopts a first crediting period of 7 years. The expected average annual emission reductions are 40,803 tCO₂e/y. Accordingly, the project is expected to generate 264,193 tCO₂e emissions reduction throughout its first crediting period.

The validation team confirmed that emission reduction forecast is reasonable if the underlying assumptions do not change.

The validation team confirms that the project activity falls under the category 1 'Project' as per para 3.10 of the VCS Standard version 4.4, as it has potential to reduce GHG emission less than 300,000 tCO₂e/year. The spatial extent of project boundary is till grid. The project activity will supply electricity to the Grid through transmission lines connected through sub-stations.

Since the project activity generates electricity through Landfill Gas, a clean renewable energy source it doesn't cause any negative impact on the environment and thereby contributes to climate change mitigation efforts.

Validation team confirmed that the project activity is not a grouped project.

The project activity is in compliance to Related laws and regulations to the proposed project activity are as followed:

- a. Electricity Market Law (³Law Number: 6446, ratified 14/03/2013, enacted 30/03/2013)
- b. Law on Utilization of Renewable Energy Resources for the Purpose of Generating Electricity Energy⁴ [Law Number: 5346 Ratification Date: 10.05.2005 Enactment Date: 18.05.2005]
- c. Environmental Law⁵ [Law Number: 2827 Ratification Date: 09.08.1983 Enactment Date: 11.08.1983]
 - i. Regulation on Solid Waste Control⁶ [Last updated on 05.04.2005]
 - ii. Regulation on Managed Waste Land Filling⁷ [Regulation number 27533 Enactment Date: 26.03.2010]

Assessment team during the desk review and on-site assessment confirms that in accordance with methodology AMS-III.G Landfill methane recovery ---version 10.0 and AMS-I.D Grid connected renewable electricity generation Version 18.0 the project activity is a LFG power generation project, utilizing landfill gas consists of methane.

The scenario existing prior to the start of the implementation of the project activity is:

- LFG from Sinop Solid Waste Landfill Facility is released into the atmosphere.
- Equivalent electricity generated by the project is supplied by the grid, which is dominated by fossil fuel-based power plants.

The project ownership is the legal right to control and operate the project activities. Sinop İnşaat Sanayi ve Ticaret Limited Şirketi is the project proponent (PP) of project activity and they have the legal right to control and operate the project activities. Same has been confirmed through Generation license and EIA positive certificate. Thus accepted.

³ <https://www.mevzuat.gov.tr/mevzuat?MevzuatNo=5346&MevzuatTur=1&MevzuatTertip=5>

⁴ <https://www.resmigazete.gov.tr/eskiler/2005/05/20050518-1.htm>

⁵ <https://www.mevzuat.gov.tr/mevzuatmetin/1.5.2872.pdf>

⁶ Reference: The official web page of the Ministry of Environment and Forestry; (Website only accessible in Turkey as per local laws) <http://www.cygm.gov.tr/CYGM/Files/mevzuat/yonetmelik/kaky.doc>

⁷ <https://www.resmigazete.gov.tr/eskiler/2010/03/20100326-13.htm>

Assessment team checked the declaration provided by the PP and confirmed that the project has not been registered or is seeking registration under any other GHG programs i.e., CDM⁸, GS⁹ & GCC¹⁰. Further, Assessment team confirms that given project activity is not a supply chain project. Hence, scope 3 emissions are not applicable.

Assessment team checked onsite and confirms that the details of the project proponent areas below

Organization name	Biokütle Enerji Elektrik Üretim Anonim Şirketi
Contact person	Savaş Uzun
Title	CEO
Address	Soğuksu Mah. 308 Sok. Adabel Sit. No:42/5 Muratpaşa, Antalya
Telephone	+90 (242) 237 31 00
Email	savas.uzun@biosenerji.com

Assessment team also checked the details of other entity and found correct. The details are as below:

Organization name	Life İklim ve Enerji Ltd. Şti.
Role in the project	Carbon Certification Consultant
Contact person	Doğukan Arıkan
Title	Expert
Address	Oğuzlar Mah. 1377. Sk. No:19/9 06520, Balgat, Çankaya/Ankara
Telephone	+90 312 481 2142
Email	dogukan.arikan@lifeenerji.com

First gas engine of 1.453 MWm/ 1.414 MWe is commissioned on 29-June-2021, which has been confirmed through Provisional acceptance letter issued by Government of Turkey. However, in accordance with the electricity generation records, Project started generating electricity from 01-July-2021, thus this date has been considered as project start date. PP has chosen 07-year renewable crediting period. The

⁸ <https://cdm.unfccc.int/Projects/projsearch.html>

⁹ <https://registry.goldstandard.org/projects?q=&page=1>

¹⁰ <https://projects.globalcarboncouncil.com/>

1st crediting period was 01-July-2021 to 30-June-2028. It was confirmed that current Joint PD & MR is being developed under the first crediting period which will start from 01-July-2021.

The project activity is not a part of grouped project.

No commercially sensitive information has been excluded from the public version of the project description.

Estimated average annual GHG emission reductions from the project activity are 40,803tCO₂e per year, which is less than 300,000 tCO₂e per year. Hence, the activity are classified as “Project”.

Project Scale	
Project	✓
Large project	

The estimated GHG Emission Reductions will depend upon the actual energy supplied to grid by the project activity. The tentative Emission Reduction for the project activity are as follows:

Year	Estimated GHG emission reductions or removals (tCO ₂ e)
01-July-2021 – 31-December-2021	12,979
01-January-2022 – 31-December-2022	29,743
01-January-2023 – 31-December-2023	33,186
01-January-2024 – 31-December-2024	42,173
01-January-2025 – 31-December-2025	44,917
01-January-2026 – 31-December-2026	47,364
01-January-2027 – 31-December-2027	49,562
01-January-2028 – 30-June-2028	25,703
Total estimated ERs	285,627
Total number of crediting years	7
Average annual ERs	40,803

No commercially sensitive information has been excluded from the public version of the project description. The details are presented transparently to the assessment team for analysis which lead to positive conclusion for this validation.

The project proponent believes that the project activity has contributed to sustainable development in terms of the four indicators as follows:

Further, The Project is not rejected by other GHG programs like CDM¹¹, GCC¹² and GS¹³. A declaration for the same is checked and found correct by the assessment team.

The project promotes sustainable development through the following aspects:

Sinop Landfill Gas to Power Generation Project is located at Sinop Landfill Site in Hacıoglu Neighborhood, Central District, Sinop Province Turkey. whose objective is to collect the landfill gas (LFG) from Sinop landfill site and utilize it to generate electricity. The project involves the installation of LFG collection, pre-treatment system, with subsequent electricity generation and grid connection system.

Social wellbeing: Setting up of the project activity has improved standard of living of local peoples, Infrastructure, created new job openings and thus reduce migration hence is supporting employment of the local people around.

Environmental wellbeing: Since the project activity leads to cleaner production of energy, it is leading to displacement of fossil fuel-based energy. Since the project activity leads to lesser GHG emission it results in better air quality. Further, contributing to the climate change fight by reducing CH₄ emissions.

- Constituting a new, clean and efficient technology model for the disposal and handling of waste.
- Generating and dispatching electricity from a renewable and sustainable energy source to a grid nowadays reliant on fossil.

Assessment team confirmed that the project makes positive contributions to at least three Sustainable Development Goals (SDGs). These are:

SDG 7: Affordable and Clean Energy

(7.2.1 Renewable energy share in the total final energy consumption)

The Project is a waste to power project that generates renewable energy by capturing methane from municipal waste and utilizing it to produce electric energy through gas engine systems. By supplying renewable energy generated at the plant to the national grid, the Project contributes to increasing the share of renewable energy in the global energy mix and the proportion of the population with primary reliance on clean fuels and technology. During current monitoring 6462.01 MWh of renewable electricity to the grid.

SDG 8: Decent Work and Economic Growth

¹¹ <http://cdm.unfccc.int/>

¹² <https://www.globalcarboncouncil.com/framework/gcc-framework/>

¹³ <http://www.goldstandard.org/>

The demand for food and electric energy is rapidly increasing in Turkey for various reasons, such as industrialization, urbanization, economic development, and population growth. The country's external dependence on agricultural products has intensified because the increasing demand cannot be met by a decreasing domestic agricultural production capacity which is due to several reasons, such as shrinkage of agricultural land, increasing migration to urban spaces from rural areas where agricultural production is densely located, and dramatic increase in the cost of inputs for agricultural production. Therewithal, Turkey, which cannot meet its increasing electricity demand due to the deprivation of conventional resources used to generate energy, such as coal, oil, and natural gas, has also become foreign-dependent on energy. Through its implementation, the project activity contributes to reducing Turkey's foreign dependency by generating renewable energy out of municipal wastes. In addition, as an LFG-based renewable energy technology implementation, the project activity achieves higher levels of economic productivity; hence, increasing the annual growth rate of real GDP per employed person. Moreover, it increases the region's employment capacity while decreasing the unemployment rate.

SDG 13: Climate Action

The project activity reduces GHG emissions by capturing and utilizing methane, one of the most potent GHGs triggering climate change. It is estimated that the average annual emission reduction that the Project will generate is around 40,803 tCO₂e.

Thus, the project causes no negative impact on the surrounding environment contributing to environmental well-being. CL02, CL003, CL05, CL07 and CAR01 was raised on this section and closed successfully. Please refer Appendix 2 for further details.

3.2 Participation under Other GHG Programs

The project was not registered under any other registry. Audit team checked the I-REC Mechanism database and GS4GG project database and found that the project activity is neither registered under I-REC mechanism nor under GS scheme. Assessment team checked the following registries to confirm the same.

1. <https://www.irecstandard.org/registries/>
2. <http://cdm.unfccc.int/>
3. <http://www.goldstandard.org/>
4. <https://www.globalcarboncouncil.com/>

Rejection by other GHG program: -

The Project is not rejected by other GHG programs. A declaration for the same is checked and found correct by the assessment team. Also, assessment team checked the following registries to confirm the same. The details of the registries checked are as follows:

1. <http://cdm.unfccc.int/>
2. <http://www.goldstandard.org/>

3. <https://verra.org/>
4. <https://www.globalcarboncouncil.com/>

Renewable energy certificates are available for trading in the host country. However, the same is not availed by the project participant. The undertaking regarding the same is submitted by PP which is acceptable to the assessment team and assessment team also checked and found the declaration to be correct. Please refer Appendix 2 for further details. Further, Assessment team confirms that given project activity is not a supply chain project. Hence, scope 3 emissions are not applicable.

Assessment team conclude that the project activity is eligible to participate under VCS program as per eligibility criterion given under VCS standard version 4.4.

3.3 Safeguards

3.3.1 No Net Harm

Water Quality

The amount of wastewater which is produced by the project is very small. Wastewater is originated from leachate at the landfill. In order to prevent this pollution hazard, first, base isolation was made in the existing SWDS and leakage from the SWDS to the soil and mixing with groundwater was prevented. Secondly, with the leachate drainage system installed at each stage, the leachate is collected and pumped back to the lot area at certain periods, so that both some of it is evaporate and the gas yield is increased.

Air Quality

This project uses LFG which is collected from Sinop Landfill site to generate electricity, and it avoids uncontrolled releasing of LFG. Therefore, it reduces greenhouse gas and effluvium emitting to air, and mitigates the possible danger of fire or explosion. The emission of NO_x, SO₂ and NH₃ for this project is less than the emission standards, that is because prior to electricity generation, LFG has been pre-treated to remove its impurities and moisture. After that, LFG is sent to gas-fired engines. And the exhaust gas would be emitted to atmosphere which is far away from sensitive targets. Therefore, the emission of exhaust gas has little impact on surrounding environment.

Soil condition

This project does not produce any chemicals or hazardous waste (NO_x, SO_x, VOC, mercury, or cadmium) thus; it does not pollute the soil condition in quality and quantity. So, Sinop Landfill Gas to Electricity Power Plant Project does not have negative impact on soil condition compared to the baseline.

Noise

The gas engine is housed in the setup with noise attenuation facilities thus ruling out any impact of noise. The nearest settlement to the project is the Hurdacılar block with 300 m distance. The noise of the project has no impact on surrounding residential areas.

Solid Waste

The project includes capture and utilisation of methane emanating from the landfilling operation and does not result in production for solid waste.

Quality of employment

This project provides local employment during the construction and operational phase. The employees are trained on health and security issues. All the trainings have been provided in accordance with "Regulations of Ministry Labor and Social Security". Trainings and certificates also increase their capacity to be able to work in such an environment.

Livelihood of the poor

This project creates job opportunities especially to young people and gives the opportunity to provide a living to their families without trying to look for a job in bigger cities for the employees. The social security system through a contract is also changing living standards of the local people.

Access to affordable and clean energy services

Dependency of Turkish electricity generation to imported fuels is projected to remain too high. So, any electricity generation from natural and local sources (waste to energy) decreases Turkey's dependency of imported fuels. Therefore, from local sources decreases Turkey's dependency of imported fuels, helps to attain more sustainable and affordable electricity service.

Balance of payments and investment

Natural gas is used domestically mainly in the electric power sector. This project supports to decrease the dependency of fuel import and result in net foreign currency saving and energy security. Thus, this parameter is also positive, but this is not monitored.

Technology transfer and technological self-reliance

This project has positive effect on technology transfer and technological self-reliance indicator. Thus, this parameter is also positive.

Based on the information provided, Assessment team conclude that the project activity does not possess any potential negative environmental and socio-economic impacts. Furthermore, the project has taken sufficient steps to implement mitigation actions, ensuring that any potential adverse effects are effectively addressed and minimized.

3.3.2 Local Stakeholder Consultation

As per the VCS requirements, it is necessary to invite the relevant stakeholders, prior of the validation process. The assessment team checked the relevance of the dates during the validation site visit. Local stakeholder consultation was conducted through distributing questionnaires to local stakeholders on 18-October-2022(Photos of have been shared of meeting) by the project owner. The survey was conducted through distributing and collecting responses of questionnaires. The questionnaire was reasonably designed to assess the project impacts on the local environment and social economic development. Questionnaires were returned with a 100% response rate. By checking the questionnaires, the assessment team confirm that the local stakeholder has no negative comments for the construction of the project activity. Furthermore, through site interview, the assessment team confirmed that communications with stakeholders were conducted at periodic intervals confirmed with grievance register. Key implementation schedules or changes of the project has been and will be communicated to the local authority, who will inform the neighborhood committee and the local residents. The comments and suggestions from residents will be collected by the local authority and feedback to project owner.

Local government agencies and authorities will conduct spot checks on the implementation of the project, and give suggestions to the project owner on the involved rectification problems. By checking the filled questionnaires from local stakeholders, the assessment team confirmed that communications with local stakeholders was being carried out in 18-October-2022 by the project owner during this monitoring period. There are no negative comments received for the project during this monitoring period. Furthermore, via interviewing with staff from local Ecology and Environment Experts and local residents during the site visit, it is confirmed by the assessment team that during the implementation stage of this monitoring period, local authority has conducted follow-up interviews and ongoing communication with local stakeholders to collect any comments or inputs to the project implementation.

Assessment team verified the documents submitted by the PP like attendance sheet, invitation letter and feedback provided by the local stakeholder. It was confirmed that PP has placed grievance register at the project site which provides stakeholder opportunity to report any grievance they have. Thus, assessment team is of opinion that the local stakeholder consultation is adequate and appropriate.

3.3.3 Environmental Impact

An Environmental Impact Assessment Positive Certificate has been provided by the project participants. The Law defines among others the general requirements of the Environmental Impact Assessment (EIA) states specific requirements according to the type of project, location, size and other characteristics. During desk review verification team observed that the Environmental Impact Assessment of the related project, "The Sinop Landfill Gas to Electricity Power Plant Project" made by the Sinop Provincial Environmental Services Association, issuance - 30047206-220.03-15758 by the Ministry, dated 23-December-2020 is Positive. In order to generate electrical energy from the gases to be collected from the solid waste stored on the land, whose coordinates have been given in the attached list and the EIA Positive Decision has been taken, "Production of Electrical Energy from the Methane Gas Coming from the Storage Area in the "Sinop Landfill Gas to Electricity Power Plant (2,906 MWm/2,828 MWe)" project was evaluated as out of scope as it was less than the threshold value in the EIA Regulation Lists published in the Official Gazette dated 25-November-2014 and numbered 29186, and an additional report was prepared and added to the EIA Positive Decision in the "e-ÇED system".

Considering the fact, that the project activity already obtained the mandatory regulation, it is therefore possible to claim that there is no net environmental impact linked to the project.

3.3.4 Public Comments

The Global Stakeholder Consultation commenting period had been completed before first crediting period. The assessment team noted that this project was open for public comment from 12-January-2023 to 11-February-2023. The detail was checked by the assessment team in the following web platform:

<https://registry.verra.org/app/projectDetail/VCS/4014>

During the period, no public comments were received.

3.3.5 AFOLU-Specific Safeguards

The project activity is not an AFOLU project.

For non-AFOLU projects, this section is not required.

3.4 Application of Methodology

3.4.1 Title and Reference

The following approved baseline & monitoring methodologies are applied:

1. **AMS-III.G** Small-scale Methodology /3/: Landfill methane recovery. Version 10.0,
2. **AMS-I. D** Grid connected renewable electricity generation -- Version 18.0¹⁴

Methodological Tool /4/ implemented:

- Tool 05-Tool to calculate baseline, project and/or leakage emissions from electricity consumption Version 03.0¹⁵.
- Tool 04-Methodological Tool: Emissions from solid waste disposal sites, Version 08.1¹⁶.
- Tool 32-Methodological Tool: Positive lists of technologies. Version 04.0¹⁷
- Tool 06-Methodological Tool: Project emissions from flaring. Version 04.0
- Tool 07: “Tool to calculate the emission factor for an electricity system” Version 07.0¹⁸
- Methodological Tool: Tool 08: “Tool to determine the mass flow of a greenhouse gas in a gaseous stream” Version 03.0¹⁹

3.4.2 Applicability

The project activity involves waste management technologies including LFG extraction, and utilization of the produced LFG for electricity generation. The applicability of the methodologies and Tool /4/ have been discussed below. The applicability of the project is validated against the current version of the methodology /3/.

Applicability of AMS-III.G (Version.10)

For the landfill gas recovery component of the proposed VER project, the emission reductions resulting from the capture and utilization of landfill gas are calculated using AMS-III.G “Landfill methane recovery” Version 10.0. This methodology /3/ is applicable to landfill gas capture project activities, where the baseline scenario is the partial or total atmospheric release of the gas, and the project activities include activities such as:

Ref	Applicability Criteria	Justification	Assessment
AMS-III.G	Different options to utilize the recovered landfill gas as detailed in paragraph 4 of “AMS-III.H.: Methane recovery in wastewater treatment” (version 19.0) are eligible for use under this	The project includes capture and utilization of the landfill gas for electrical energy generation, which complies to 4(a) of para.4 of AMS-III.H.	By checking feasibility report and provisional acceptance letter during desk review and site visit interview and

¹⁴ <https://cdm.unfccc.int/methodologies/DB/W3TINZ7KKWCK7L8WTXFQQOFQQH4SBK>

¹⁵ <https://cdm.unfccc.int/methodologies/PAmethodologies/Tool/4/s/am-Tool/4/-05-v3.0.pdf>

¹⁶ <https://cdm.unfccc.int/methodologies/PAmethodologies/Tool/4/s/am-Tool/4/-04-v8.1.pdf>

¹⁷ <https://cdm.unfccc.int/methodologies/PAmethodologies/Tool/4/s/am-Tool/4/-32-v4.0.pdf>

¹⁸ <https://cdm.unfccc.int/methodologies/PAmethodologies/Tool/4/s/am-Tool/4/-07-v7.0.pdf>

¹⁹ <https://cdm.unfccc.int/methodologies/PAmethodologies/Tool/4/s/am-Tool/4/-08-v3.0.pdf>

	methodology /3/. The relevant procedures in AMS- III.H. shall be followed in this regard. AMS-III.H (version 19.0): paragraph 4 states that the recovered biogas from the above measures may also be utilized for the following applications instead of combustion/flaring: (a) Thermal or mechanical, electrical energy generation directly. (b) Thermal or mechanical, electrical energy generation after bottling of upgraded biogas, in this case additional guidance provided in the appendix shall be followed; or (c) Thermal or mechanical, electrical energy generation after upgrading and distribution, in this case additional guidance provided in the appendix shall be followed: (i) Upgrading and injection of biogas into a natural gas distribution grid with no significant transmission constraints; (ii) Upgrading and transportation of biogas via a dedicated piped network to a group of end users; or (iii) Upgrading and transportation of biogas (e.g., by trucks) to distribution points for end users; (d) Hydrogen production; (e) Use as fuel in transportation applications after upgrading. AMS-III. H: paragraph 5 states that If the recovered biogas is	the project activity complies to the options for landfill gas recovery and utilization as detailed in paragraph 4 of “AMS-III.H.: Methane recovery in wastewater treatment” (version 19.0) therefore the applicability of the methodology /3/ AMS-III.G (version 19.0) is established.	it is confirmed that the project utilizes the recovered LFG to generate electrical energy directly which meet the requirement of 4 (a) of AMS-III.H. (version 19.0).
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	used for project activities covered under paragraph 4(a), that component of the project activity can use a corresponding methodology /3/ under Type I.		
	Measures are limited to those that result in aggregate emission reductions of less than or equal to 60 kt CO ₂ equivalent annually from all Type	The avoided emission from the project activity is 40,803 tCO ₂ /annum which is less than 60 kt CO ₂ e, hence methodology /3/ AMS-III.G. is applicable to the project activity.	It is confirmed from submitted ER sheet the annual ER is 40,803 tCO ₂ /annum of the project is less than 60 kt CO ₂ e from all type III components.
	The proposed project activity does not reduce the amount of organic waste that would have been recycled in the absence of the project activity.	The project does not reduce the amount of organic waste that would have been recycled in the absence of the project activity as there are no recycling facilities available to which the municipal waste would have been directed in absence of the project activity. Hence methodology /3/ AMS-III.G. is applicable to the project activity.	By checking feasibility report during desk review and site visit interview, it is confirmed that the implementation of the project does not reduce the amount of organic waste that would be recycled in the absence of the project. All the solid waste is disposed in the Sinop Landfill site.
	This methodology /3/ is not applicable if the management of the solid waste disposal site (SWDS) in the project activity is deliberately changed in order to increase methane generation compared to the situation prior to the implementation of the project activity (e.g., other than to meet a technical or regulatory requirement). Such changes may include, for example, the addition of liquids to a SWDS, pre- treating	The management of the solid waste disposal site (SWDS) in the project activity is not deliberately changed in order to increase methane generation compared to the situation prior to the implementation of the project activity. Hence methodology /3/ AMS-III.G. is applicable to the project activity.	By checking feasibility report and provisional acceptance letter during desk review and site visit interview, it is confirmed that this criterion is not applicable for the project.

	waste to seed it with bacteria for the purpose of increasing the rate of anaerobic degradation of the SWDS or changing the shape of the SWDS to increase methane production.		
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Ref	Applicability Criteria	Justification	Assessment
AMS-I.D	This methodology /3/ is applicable to project activities that: (a) Install a Greenfield plant; (b) Involve a capacity addition in (an) existing plant(s); (c) Involve a retrofit of (an) existing plant(s); (d) Involve a rehabilitation of (an) existing plant(s)/unit(s); or Involve a replacement of (an) existing plant(s).	The project installs a Greenfield LFG power plant at the existing Sinop Landfill where there was no renewable energy power plant operating prior to the implementation of the project activity. Thus, the project activity includes installation of greenfield power project. Applicable.	By checking provisional acceptance letter during desk review and site visit interview, it is confirmed that the project is a greenfield project.
	Hydro power plants with reservoirs that satisfy at least one of the following conditions are eligible to apply this methodology /3/: (a) The project activity is implemented in an existing reservoir with no change in the volume of reservoir; (b) The project activity is implemented in an existing reservoir, where the volume of reservoir is increased and the power density of the project activity, as per definitions given in the project emissions section, is greater than 4 W/ m² (c) The project activity results in new reservoirs and the	The project activity includes implementation of landfill gas extraction and utilization unit for the purpose of power generation and does not include hydro power plant therefore the applicability condition is not applicable/relevant to the project activity as the applicability conditions is related to hydro power projects. Not Applicable.	By checking provisional acceptance letter during desk review and site visit interview, it is confirmed this is not applicable as project is not a hydro power project.

	power density of the power plant, as per definitions given in the project emissions section, is greater than 4 W/ m ² .		
	If the new unit has both renewable and non-renewable components (e.g., a wind/diesel unit), the eligibility limit of 15 MW for a small scale CDM project activity applies only to the renewable component. If the new unit co fires fossil fuel, the capacity of the entire unit shall not exceed the limit of 15 MW.	The project activity includes renewable component only i.e., LFG based power generation unit of 2.828 MWe. Therefore, the applicability condition is not applicable/relevant to the project activity.	By checking Generation license during desk review and site visit interview, it is confirmed the total installed capacity of the project is 2.828 MWe.
	Combined heat and power (co-generation) systems are not eligible under this category.	The project activity includes LFG based power generation unit and not a combined heat and power (co-generation) systems therefore the applicability condition is not applicable/relevant to the project activity as the applicability conditions is related to co-generation unit.	By checking provisional acceptance letter & Generation License during desk review and site visit interview, it is confirmed the project does not involve in heat generation.
	In the case of project activities that involve the capacity addition of renewable energy generation units at an existing renewable power generation facility, the added capacity of the units added by the project should be lower than 15 MW and should be physically distinct ¹ from the existing units.	The project does not involve the capacity addition at an existing renewable power generation facility yet. If it happens soon, it's still going to be lower 15 MW and be physically distinct. Hence applicability condition is not applicable/relevant to the project activity.	By checking provisional acceptance letter & Generation License during desk review and site visit interview, it is confirmed the project does not involve in capacity addition.
	In the case of retrofit, rehabilitation, or replacement, to qualify as a small-scale project, the total output of the retrofitted, rehabilitated or replacement	The project activity is a greenfield installation of LFG extraction and LFG based power generation unit of 2.828 MWe and does not	By checking provisional acceptance letter & Generation License and site visit

	power plant/unit shall not exceed the limit of 15 MW.	include retrofit, rehabilitation or replacement and hence applicability condition is not applicable/relevant to the project activity.	interview, it is confirmed the project does not involve in retrofit, rehabilitation or replacement.
	In the case of landfill gas, waste gas, wastewater treatment and agro-industries projects, recovered methane emissions are eligible under a relevant Type III category. If the recovered methane is used for electricity generation for supply to a grid, then the baseline for the electricity component shall be in accordance with procedure prescribed under this methodology /3/. If the recovered methane is used for heat generation or cogeneration other applicable Type-I methodologies such as “AMS- I.C.: Thermal energy production with or without electricity” shall be explored.	The project recovers LFG for electricity generation. Hence recovered methane emissions are eligible under Type III methodology /3/ AMS-III.G. And the baseline for the electricity component is in accordance with procedure prescribed in AMS-I.D. Applicable.	By checking provisional acceptance letter, Feasibility report & Generation License and site visit interview, it is confirmed the project is designed to install four engines to combust LFG of Sinop MSW, which mainly contains 50% methane.
	In case biomass is sourced from dedicated plantations, the applicability criteria in the Tool /4/ “Project emissions from cultivation of biomass” shall apply.	The project does not involve biomass sourced from dedicated plantations. Hence the applicability condition is not applicable/relevant to the project activity.	By checking feasibility report during desk review and site visit interview, it is confirmed that the project does not involve in biomass.

The baseline scenario of the proposed project is total atmospheric release of the gas (see section B.4) and the captured gas is used to produce energy. Thus, the VER project activity corresponds to **AMS-III.G** Small-scale Methodology /3/: Landfill methane recovery. Version 10.0, **AMS-I. D** Grid connected renewable electricity generation --- Version 18.0, is applicable to the project activity.

Applicability of “Project emissions from flaring” TOOL /4/06 Version 04.0

In line with **AMS-III.G** Small-scale Methodology /3/: Landfill methane recovery. Version 10.0, and **AMS-I.D** Grid connected renewable electricity generation Version 18.0, the project activity is eligible to claim emission reductions from its flaring activities. Within the project activity, there is

an gas ballon system installed to collect any excess of LFG. In accordance with the above explanation, the “Project emissions from flaring” version 04.0, is not applied to the proposed project activity and no emission reductions will be claimed from the flaring activity throughout the crediting period.

Applicability of “Emissions from solid waste disposal sites” TOOL /4/04 Version 08.1

The Tool /4/ provides procedures to calculate baseline, project or leakage emissions of methane from solid waste disposed or prevented from disposal at a solid waste disposal site (SWDS). The Tool /4/ is applicable as the solid waste disposal site where the waste would be dumped can be clearly identified and the proposed project is not a stockpile, and no hazardous waste is processed.

The applicability of the Tool /4/ is discussed as follows:

Ref	Applicability Criteria	Justification	Assessment
Application A	The CDM project activity mitigates methane emissions from a specific existing SWDS.	Since the methane emissions are mitigated by capturing and flaring or combusting the methane, “Application A” implies to the proposed project activity. The methane is generated from waste disposed in the past, including prior to the start of the CDM project activity. In line with the requirements, this Tool /4/ is only applied for an ex-ante estimation of emissions in the CDM-PDD. The emissions will be monitored during the crediting period using the applicable approaches in the relevant methodologies (AMS-III.G and AMS-I.D in this case).	During assessment, Verification team found that this project activity mitigates methane emission from existing SWDS of Sinop. This is verified during onsite visit and desk review of provided PD & MR and PDD. Hence Verification team concluded that “Emissions from solid waste disposal sites” TOOL /4/04 Version 08.1 section A is applicable for this project activity.
Application B	The CDM project activity avoids or involves the	Since the MSW is not treated with an alternative option such as composting or anaerobic digestion, “Application B” is not relevant	During assessment, Verification team found that this project activity does not involves the disposal of waste at SWDS

Ref	Applicability Criteria	Justification	Assessment
	disposal of waste at a SWDS.	to the proposed project activity.	of Sinop. This is verified during onsite visit and desk review of provided PD & MR and PDD. Hence Verification team concluded that "Emissions from solid waste disposal sites" TOOL /4/ 04 Version 08.1 section B is not applicable for this project activity.

The Tool /4/ "Emissions from solid waste disposal sites" TOOL /4/04 version 08.1 is applicable since the proposed project activity mitigates methane emissions from a specific existing SWDS.

Applicability of "Tool to calculate the emission factor for an electricity system" TOOL 07 Version 07.0

This methodological Tool /4/ determines the CO₂ emission factor for displacement of electricity generated by power plants in an electricity system, by calculating the "combined margin" emission factor (CM) of the electricity system. This Tool /4/ maybe applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity. As the proposed project activity supplies electricity to the national grid, the "Tool /4/ to calculate the emission factor for an electricity system" version 07.0 is applicable to the project activity.

Applicability of "Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation" TOOL 05 version 03.0:

These Tool /4/ provides procedures to estimate the baseline, project and/or leakage emissions associated with the consumption of electricity. The Tool /4/ is only applicable if certain scenario applies to the sources of electricity consumption. Three possible scenarios have been discussed as follows:

Ref	Applicability Criteria	Justification	Assessment
A	If emissions are calculated for electricity consumption, the Tool /4/ is only applicable if one out of the following three scenarios	In the proposed project activity, the electricity is purchased from the grid only and no captive power plant is installed at the site	During assessment, verification team observed that this project involves scenario A electricity consumption from grid and no captive

Ref	Applicability Criteria	Justification	Assessment
	applies to the sources of electricity consumption: (a) Scenario A: Electricity consumption from the grid. The electricity is purchased from the grid only, and either no captive power plant(s) is/are installed at the site of electricity consumption or, if any captive power plant exists on site, it is either not operating or it is not physically able to provide electricity to the electricity consumer; (b) Scenario B: Electricity consumption from (an) off-grid fossil fuel fired captive power plant(s). One or more fossil fuel fired captive power plants are installed at the site of the electricity consumer and supply the consumer with electricity. The captive power plant(s) is/are not connected to the electricity grid; or (c) Scenario C: Electricity consumption from the grid and (a) fossil fuel fired captive power plant(s). One or more fossil fuel fired captive power plants operate at the site of the electricity consumer. The captive	of electricity consumption (Scenario A).	power plant is involved in this project activity and accepted by verification team during desk review and onsite visit of auditing team.

Ref	Applicability Criteria	Justification	Assessment
	power plant(s) can provide electricity to the electricity consumer. The captive power plant(s) is/are also connected to the electricity grid. Hence, the electricity consumer can be provided with electricity from the captive power plant(s) and the grid.		
B	This Tool /4/ can be referred to in methodologies to provide procedures to monitor amount of electricity generated in the project scenario, only if one out of the following three project scenarios applies to the recipient of the electricity generated: (a) Scenario I: Electricity is supplied to the grid; (b) Scenario II: Electricity is supplied to consumers/electricity consuming facilities; or (c) Scenario III: Electricity is supplied to the grid and consumers/electricity consuming facilities	The projects entail compliance with Scenario I: Electricity is supplied to the grid.	During assessment, verification team observed that this project involves scenario I electricity supply to the grid and not used for captive consumption in this project activity and accepted by verification team during desk review and onsite visit of auditing team.
C	This Tool /4/ is not applicable in cases where captive renewable power generation technologies are installed to provide electricity in the project	No captive renewable power generation technologies are installed to provide electricity in the project activity, in the	During assessment, verification team observed that this project do not involves scenario of captive energy consumption of power is

Ref	Applicability Criteria	Justification	Assessment
	activity, in the baseline scenario or to sources of leakage. The Tool /4/ only accounts for CO ₂ emissions.	baseline scenario or to sources of leakage.	involved in this project activity and accepted by verification team during desk review and onsite visit of auditing team.

Based on the above arguments it can be concluded that the “Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation’ version 03.0 is applicable to the project activity.

Applicability of “Tool to determine the mass flow of a greenhouse gas in a gaseous stream” Version 03.0.0:

These Tool /4/ provides procedures to determine the mass flow of CH₄ in the LFG in time interval “t”. As referred in methodology /3/ ACM0001, the “Tool /4/ to determine the mass flow of a greenhouse gas in a gaseous stream” version 03.0.0 is applicable for the proposed project activity.

Applicability of “Tool 32 Positive lists of technologies” - Version 04.0:

The project activity adopts this methodological tool for demonstrating their additionality. This methodological tool shall be applied in conjunction with a small-scale or large-scale methodology which refers to this tool. This methodological tool is applied in conjunction with AMS-III.G which refers to this tool. The positive lists as contained in section 5 of this tool are valid up to 10/03/2025. Notwithstanding the provisions on the validity of new, revised and previous versions of methodologies and methodological tools in the “Procedure: Development, revision and clarification of baseline and monitoring methodologies and methodological tools”, there will be no grace period for the application of this tool and the validity of the positive list after this date, including in cases where further technologies are added to the positive list through revisions of this tool before this date. The positive lists as contained in section 5 “5.1.1. Landfill gas recovery and its gainful use” of this tool are valid up to 10/03/2025. It is confirmed by the assessment team that the application of the baseline methodology is transparent and conservative and confirms that the chosen baseline and monitoring methodology i.e. AMS-III.G version 10.0 and AMS-I.D version 18.0 are applicable to the project activity. The project activity qualifies as Type I and Type III during every year of the crediting period in accordance with applicable provisions for project activity eligibility as discussed above. Also, annual average GHG emission reductions or removal per year of the project activity is 40,803 tCO₂e, less than 60,000 tonnes of CO₂e per year, which is applicable as per small scale project activities methodology AMS-III.G version 10.0 and AMS-I.D version 18.0.

3.4.3 Project Boundary

The spatial boundary includes the power plant/unit and all power plants/units connected physically to the electricity system that the project power plant is connected to as verified from the applied methodology /3/ AMS-III.G and AMS-I.D. VVB could confirm this during the onsite audit, also from the commissioning certificates^{2/}. The assessment team was able to confirm that all the identified emission sources which are impacted by the project activity are addressed by the approved methodology /3/ and can be seen in the Table below;

Source		Gas	Included?	Justification/Explanation
Baseline	Emissions from decomposition of waste at the SWDS site	CO ₂	No	CO ₂ emissions from the decomposition of organic waste are not accounted for since the CO ₂ is also released under the project activity
		CH ₄	Yes	Major source of emissions in the baseline.
		N ₂ O	No	N ₂ O emissions are small compared to CH ₄ emissions from SWDS. This is conservative
		Other	No	Excluded for simplification. This is conservative
	Emissions from electricity generation	CO ₂	Yes	Major emission source if power generation is included in the project activity
		CH ₄	No	Excluded for simplification. This is conservative
		N ₂ O	No	Excluded for simplification. This is conservative
		Other		
Project	Emissions from electricity consumption due to the Project activity	CO ₂	Yes	May be an important emission source.
		CH ₄	No	excluded for simplification. This emission source is assumed to be very small
		N ₂ O	No	excluded for simplification. This emission source is assumed to be very small
		Other	No	Excluded for simplification. This is conservative
	LFG capture and flaring	CO ₂	No	It is part of the natural carbon cycle, where the carbon was previously sequestered from the atmosphere.
		CH ₄	Yes	Excluded for simplification. This is conservative.

Source	Gas	Included?	Justification/Explanation
	N ₂ O	No	Excluded for simplification. This is conservative.
	Other	No	Excluded for simplification. This is conservative.

The assessment team confirms that the Joint PD & MR has correctly described the project boundary, including the physical delineation of the project activity and complies with VCS Standard version 4.4

3.4.4 Baseline Scenario

As per AMS-III.G, the baseline scenario is the situation where, in the absence of the project activity, municipal solid waste (biomass and other organic matter) is left to decay within the project boundary, and methane is emitted to the atmosphere, possibly with capture of LFG and destruction through flaring to comply with regulations or contractual requirements. No national or local safety requirements or legal regulations are in place that require methane from the landfill to be captured/ utilised and/or flared.

As per AMS-I.D, the baseline scenario is that the electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources into the grid. Since the project activity is a greenfield project activity, the baseline scenario for greenfield power plant is applicable (AMS-I.D,) to the project activity.

Therefore, the baseline scenario for the project activity is in the absence of the project activity, LFG is emitted to the atmosphere, and the electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources into the grid.

Baseline Emission Factor

Developing the baseline emission factor and calculation of the emission reductions for electricity generated under the proposed project activity is calculated according to “Tool to calculate the emission factor of an electricity system” Tool 7 version 07.0.

Emission factor has been calculated in a conservative manner as proposed by the methodology. Basic assumptions made are:

1. Emission factor remains same over the crediting period,
2. Emission factor of fuels sources is “0” or the lowest value in the references when there is no information.

According to the “Tool to calculate the emission factor for an electricity system v 07.0”, Option 1 has been selected.

The CO₂ emission factor for the displacement of electricity generated by power plants in the project activity and fed to the grid is estimated using Tool to calculate the emission factor for an electricity system version 7.0. The estimation is carried out through following steps

- a) Step 1: Identify the relevant electricity systems;

- b) Step 2: Choose whether to include off-grid power plants in the project electricity system (optional);
- c) Step 3: Select a method to determine the operating margin (OM);
- d) Step 4: Calculate the operating margin emission factor according to the selected method;
- e) Step 5: Calculate the build margin (BM) emission factor;
- f) Step 6: Calculate the combined margin (CM) emission factor.

Step 1: Identify the relevant electricity systems

As described in tool “For determining the electricity emission factors, identify the relevant project electricity system. Similarly, identify any connected electricity systems”. It also states that “If the DNA of the host country has published a delineation of the project electricity system and connected electricity systems, these delineations should be used”.

Keeping this into consideration the delineation of the national grid of Turkey by "TÜRKİYE ELEKTRİK İLETİM A.Ş. GENEL MÜDÜRLÜĞÜ" is considered for estimation.

Step 2: Choose whether to include off-grid power plants in the project electricity system (optional)

Project participants may choose between the following two options to calculate the operating margin and build margin emission factor:

Option I: Only grid power plants are included in the calculation.

Option II: Both grid power plants and off-grid power plants are included in the calculation.

The Project Participant has chosen only grid power plants in the calculation.

Step 3: Select a method to determine the operating margin (OM)

The calculation of the operating margin emission factor ($EF_{grid,OM\ simple,y}$) is based on one of the following methods, which are described under Step 4:

- a) Simple OM; or
- b) Simple adjusted OM; or
- c) Dispatch data analysis OM; or
- d) Average OM.

The data required to calculate Simple adjusted OM and Dispatch data analysis OM is not possible due to lack of availability of data to project developers. The choice of other two options for calculating operating margin emission factor depends on generation of electricity from low-cost/must-run sources. In the context of the methodology low cost/must run resources typically include hydro, geothermal, wind, low-cost biomass, nuclear and solar generation.

In accordance to Electricity Market Production Capacity Projections List²⁰

²⁰ EMRA | Energy Market Regulatory Authority (epdk.gov.tr)

Years	2016	2017	2018	2019	2020	2021	Share in 2021 (%)
Lignite	61.24	61.33	61.33	61.33	61.33	61.33	14.26%
Hard Coal	5.91	5.91	5.91	5.91	5.91	5.91	1.37%
Imported Coal	37.90	39.63	39.67	39.67	39.67	26.82	6.23%
Natural Gas	175.07	175.15	176.01	175.79	176.01	176.01	40.91%
Geothermal	1.40	1.40	1.40	1.40	1.40	1.40	0.33%
Fuel Oil	9.22	9.22	9.22	9.22	9.22	9.22	2.14%
Diesel	0.15	0.15	0.15	0.15	0.15	0.15	0.03%
Nuclear	0	0	0	4.200	12.60	21.00	4.88%
Other	1.41	1.41	1.41	1.41	1.41	1.41	0.33%
Thermal Total	292.30	294.21	295.11	299.08	307.71	316.11	70.49%
Biogas + Waste	1.20	1.20	1.20	1.20	1.20	1.20	0.28%
Hydro	104.44	112.71	115.78	116.56	116.56	116.56	27.09%
Wind	9.21	9.21	9.21	9.21	9.21	9.21	2.14%
Total	407148	417317	421292	426044	434670	443070	100.0%

The above data clearly shows that the percentage of total grid generation by low-cost/ must-run plants (on the basis of average of five most recent years) for the national grid of Turkey is less than 50 % of the total generation capacity. Thus, the Average OM method cannot be applied, as low cost/must run resources constitute less than 50% of total grid generation. The simple OM emission factor is calculated as the generation-weighted average CO₂ emissions per unit net electricity generation (tCO₂/MWh) of all generating power plants serving the system, not including low-cost/must-run power plants/units.

For the simple OM, the simple adjusted OM and the average OM, the emissions factor can be calculated using either of the two following data vintages:

- a) Ex-ante option: if the ex-ante option is chosen, the emission factor is determined once at the validation stage, thus no monitoring and recalculation of the emissions factor during the crediting period is required.

OR

- b) Ex-post option: if the ex-post option is chosen, the emission factor is determined for the year in which the project activity displaces grid electricity, requiring the emissions factor to be updated annually during monitoring.

Step 4: Calculate the operating margin emission factor

For OM factor calculation, Chronological order of power generation plants from TEİAŞ Load Dispatch Department with, fuel types, electricity generation for the calculated year were used as

input data. By using all the data which were mentioned above, Turkish Ministry of Energy and Natural Resources calculated $EF_{grid,OM,y}$ ²¹:

$$EF_{grid,OM,y} = 0.7424 \text{ tCO}_2/\text{MWh}$$

Step 5: Calculate the build margin (BM) emission factor ($EF_{grid,BM,y}$)

As per Methodological tool: "Tool to calculate the emission factor for an electricity system" (Version 07.0, EB 100, Annex 4) para 72:

In terms of vintage of data, project participants can choose between one of the following two options:

- a) Option 1 - for the first crediting period, calculate the build margin emission factor ex ante based on the most recent information available on units already built for sample group at the time of PDD submission to the VVB for project verification. For the second crediting period, the build margin emission factor should be updated based on the most recent information available on units already built at the time of submission of the request for renewal of the crediting period to the DOE. For the third crediting period, the build margin emission factor calculated for the second crediting period should be used. This option does not require monitoring the emission factor during the crediting period.
- b) Option 2 - For the first crediting period, the build margin emission factor shall be updated annually, ex post, including those units built up to the year of registration of the project activity or, if information up to the year of registration is not yet available, including those units built up to the latest year for which information is available. For the second crediting period, the build margin emissions factor shall be calculated ex ante, as described in Option 1 above. For the third crediting period, the build margin emission factor calculated for the second crediting period should be used.

Option 1 as described above is chosen by PP to calculate the build margin emission factor for the project activity. BM is calculated ex-ante based on the most recent information available at the time of submission of PD and is fixed for the entire crediting period.

For BM factor calculation by Turkish Ministry of Energy and Natural Resources $EF_{grid,BM,y}$ ²² chronological order of power generation plants from TEİAŞ Load Dispatch Department with commissioning dates, plant names, fuel types, installed power values, electricity generation for the calculated year were used as input data.

$$EF_{grid,BM,y} = 0.3680 \text{ tCO}_2/\text{MWh}$$

Step 6: Calculate the combined margin (CM) emission factor ($EF_{grid,CM,y}$)

²¹<https://enerji.gov.tr/Media/Dizin/EVCED/tr/%C3%87evreVe%C4%B0klm/%C4%B0klmDe%C4%9Fi%C5%9Fikli%C4%9Fi/TUESEmisyonFktr/Belgeler/Bform2020.pdf>

²² https://enerji.enerji.gov.tr/Media/Dizin/BHIM/tr/Duyurular/Bilgi_Formu_Web_Sitesi_2019_202110071443.pdf

As per Methodological tool: “Tool to calculate the emission factor for an electricity system” (Version 07.0, EB 100, Annex 4) para 81:

The calculation of the combined margin (CM) emission factor ($EF_{grid,CM,y}$) is based on one of the following methods:

- Weighted average CM; or
- Simplified CM.

PP has chosen option (a) i.e weighted average CM to calculate the combined margin emission factor for the project activity.

The combined margin emission factor is calculated by using weighted average CM as per TOOL07 Version 07.0 formula below:

$$EF_{grid,CM,y} = EF_{grid,OM,y} * W_{OM} + EF_{grid,BM,y} * W_{BM} \text{ ---Equation 16 in TOOL07 V7}$$

Where:

$EF_{grid,BM,y}$	Build margin CO ₂ emission factor in year y (tCO ₂ /MWh)
$EF_{grid,OM,y}$	Operating margin CO ₂ emission factor in year y (tCO ₂ /MWh)
W_{OM}	Weighting of operating margin emissions factor (%)
W_{BM}	Weighting of build margin emissions factor (%)

According to the Tool for power generation project activities other than wind and solar;

$$W_{OM} = 0.50 \text{ and } W_{BM} = 0.50$$

Then:

$$EF_{grid,CM,y} = 0.7424 \text{ tCO}_2/\text{MWh} * 0.50 + 0.3680 \text{ tCO}_2/\text{MWh} * 0.50 = 0.5552 \text{ tCO}_2/\text{MWh}$$

$$EF_{grid,CM,y} = 0.5552 \text{ tCO}_2/\text{MWh}$$

3.4.5 Additionality

The project met all Turkey laws. No further regulatory surplus need to be demonstrated. As per the Methodological Tool “Positive lists of technologies”, the project activities at new or existing landfills are deemed automatically additional, if it is demonstrated that prior to the implementation of the project activities the landfill gas (LFG) was only vented and/or flared (in the case of brownfield projects) or would have been only vented and/or flared (in the case of greenfield projects) but not utilized for energy generation, and that under the project activities any of the following conditions are met: (a) The LFG is used to generate electricity in one or several power plants with a total nameplate capacity that equals or is below 10 MW; (b) The LFG is used to generate heat for internal or external consumption; (c) The LFG is flared. The project activity uses LFG which is collected from Sinop landfill to generate electricity. Prior to the implementation of the project activity, the LFG from Sinop MSW landfill was only vented, not utilized for energy generation. By interviewing local stakeholders, staff from local government, staff from Sinop MSW landfill and checking website of local government, it is confirmed by the assessment team that only the project activity uses LFG from Sinop MSW landfill to generate electricity. By checking Project Approval

and EIA Approval, the total installed capacity of the project activity is 2.828 MWe, which is below 10 MW. Therefore, the project activity is deemed automatically additional.

3.4.6 Quantification of GHG Emission Reductions and Removals

Quantification of Baseline emissions

The proposed project activity has applied methodology /3/ AMS-III.G Version.10 and AMS-I.D Version.18 /4/. As per the paragraph 29 of the methodology /3/: “Baseline emissions are determined according to equation (1) and comprise the following sources: (a) Methane emissions from the SWDS in the absence of the project activity; (b) Electricity generation using fossil fuels or supplied by the grid in the absence of the project activity; (c) Heat generation using fossil fuels in the absence of the project activity; and (d) Natural gas used from the natural gas network in the absence of the project activity.”.

The baseline emissions are calculated as follows

$$BE_y = BE_{CH_4,y} + BE_{electricity,y} \quad (\text{Equation 1})$$

Where:

BE_y = Project Baseline emission in year y (tCO₂e)

$BE_{CH_4,y}$ = Baseline emission from LFG avoidance that otherwise would be emitted to the atmosphere in absence of the Project in year y (tCO₂e) which is estimated as per AMS-III.G

$BE_{electricity,y}$ = Baseline emission from electricity displacement that otherwise would be provided by grid in year y (tCO₂e) which is estimated as per AMS-I.D.

Baseline emissions from landfill methane recovery ($BE_{CH_4,y}$)

Baseline emissions from animal waste treatment are determined according to the applied methodology, AMS-III.G., version 10 - equation (1):

$$BE_y = \eta_{PJ} \times BE_{CH_4, SWDS,y} - (1 - OX) \times F_{CH_4,BL,y} + GWP_{CH_4} \quad (\text{Equation-2})$$

Where,

$BE_{CH_4,,}$ = Baseline emissions of methane from the SWDS in year y (t CO₂e/yr)

$BE_{CH_4,DS,y}$ = Methane emission potential of a solid waste disposal site (in t CO₂e), calculated using the methodological tool “Emissions from solid waste disposal sites”. This tool may be used:

- With the factor “f=0.0” because the amount of LFG that would have been captured and destroyed is already accounted for in this equation;
- With the definition of year x as ‘the year since the landfill started receiving wastes, x runs from the first year of landfill operation (x=1) to the year for which emissions are calculated (x=y)’.

	The amount of waste type j deposited each year x ($W_{j,x}$) shall be determined by sampling (as specified in the above-mentioned tool), in the case that waste is generated during the crediting period. Alternatively, for existing SWDS, if the pre-existing amount and composition of the wastes in the landfill are unknown, they can be estimated by using parameters related to the serviced population or industrial activity, or by comparison with other landfills with similar conditions at regional or national level.
OX	= Oxidation factor (reflecting the amount of methane from SWDS that is oxidized in the soil or other material covering the waste) (dimensionless). A default value of 0.1 may be used.
η_{PJ}	= Efficiency of the LFG capture system that will be installed in the project activity. It is used for ex ante estimation only. A default value of 50 per cent may be used.
$F_{CH_4,y}$	= Methane emissions that would be captured and destroyed to comply with national or local safety requirement or legal regulations in the year y (t CH₄). The relevant procedures in “ACM0001: Flaring or use of landfill gas” may be followed, as well as taking into account the compliance with the relevant local laws and regulation if such laws and regulations exist
GWP_{CH_4}	= Global Warming Potential for methane (value of 28).

Determination of $F_{CH_4,BL,y}$

As per ACM0001 v19.0, 4 cases given in the table below has been evaluated.

Situation at the start of the project activity	Requirement to destroy methane	Existing LFG capture and destruction system
Case 1	No	No
Case 2	Yes	No
Case 3	No	Yes
Case 4	Yes	Yes

Case 1: No requirement to destroy methane exists and no existing LFG capture system

In this situation:

$$F_{CH_4,BL,y} = 0$$

Case 2: Requirement to destroy methane exists and no existing LFG capture system

$$F_{CH_4,BL,y} = F_{CH_4,BL,R,y}$$

$F_{CH_4,BL,R,y}$ should be determined based on the information contained in the requirement to destroy methane, as follows:

(a) If the requirement specifies the amount of methane that must be flared then that amount is $F_{CH4,BL,R,y}$;

(b) If the requirement specifies a percentage of the captured LFG that is required to be flared, the amount shall be calculated as follows:

$$F_{CH4,BL,R,y} = \rho_{reg,y} \times F_{CH4,PJL,capt,y}$$

Where:

$F_{CH4,BL,R,y}$ = Amount of methane in the LFG which is flared in the baseline due to a requirement in year y (t CH₄/yr)

$\rho_{reg,y}$ = Fraction of LFG that is required to be flared due to a requirement in year y

$F_{CH4,PJL,capt,y}$ = Amount of methane in the LFG which is captured in the project activity in year y (t CH₄/yr)

$F_{CH4,PJL,capt,y}$ can be calculated by two options:

Option 1: Calculate using the “Tool to determine the mass flow of a greenhouse gas in a gaseous stream”, applying the following requirements:

- i. The gaseous stream tool shall be applied to the LFG pipeline immediately downstream of the LFG capture system and before any split in the gaseous flow to different uses or flares
- ii. CH₄ is the greenhouse gases for which the mass flow should be determined;
- iii. The simplification offered for calculating the molecular mass of the gaseous stream is valid (equations (3) or (17) in the tool); and
- iv. The mass flow should be calculated on an hourly basis for each hour h in year y.

Option 2: Calculate as the sum of the amount of methane that is sent to the flare, electricity generating or heat generating equipment in year y, however, not taking into account the working hours of the equipment;

- i. If the requirement does not specify the amount or percentage of LFG that should be destroyed but requires the installation of a capture system, without requiring the captured LFG to be flared then

$$F_{CH4,BL,R,y} = 0$$

- ii. If the requirement does not specify any amount or percentage of LFG that should be destroyed, but requires the installation of a system to capture and flare the LFG, then a typical destruction rate of 20 per cent is assumed:

$$F_{CH4,BL,R,y} = 0.2 \times F_{CH4,PJL,capt,y}$$

Case 3 and Case 4 is not evaluated since there is no existing LFG capture system in baseline and this situation will be valid always.

As a result;

$F_{CH_4,BL,y}$ is taken as zero since there is no regulation in Turkey that advises the capture of methane emissions and there is no existing LFG capture and destruction system. ($F_{CH_4,BL,y} = 0$). This was found to be acceptable by the validation team after relevant document reviews and site inspection. However, in future if regulation is changed Case 2 will be valid and $p_{reg,y}$ will be monitor accordingly during verification of emission reductions.

Parameter	Description	Unit	Value	
η_{PJ}	Efficiency of LFG capture system	%	0.85	
OX	Oxidation Factor	%	0.1	
$BE_{CH_4,y}$	Baseline Emissions associated with LFG capture	tCO ₂	2021	10,027
			2022	23,856
			2023	27,299
			2024	30,398
			2025	33,142
			2026	35,589
			2027	37,787
			2028	19,832
			Average	31,132

Determination of $BE_{CH_4,SWDS,y}$:

$BE_{CH_4,SWDS,y}$ is determined using the methodological tool "Emissions from solid waste disposal sites". The following guidance should be taken into account when applying the tool: (a) f_y in the tool shall be assigned a value of 0 because the amount of LFG that would have been captured and destroyed is already accounted for in equation (2) of methodology AMS-III.G; (b) In the tool, x begins with the year that the SWDS started receiving wastes (e.g. the first year of SWDS operation); and (c) Sampling to determine the fractions of different waste types is not necessary because the waste composition can be obtained from previous studies. For the project, the amount of methane generated from disposal of waste at the SWDS for year y ($BE_{CH_4,SWDS,y}$) is calculated using first order decay (FOD) model as follows:

$$\left. \begin{array}{l} BE_{CH_4,SWDS,y} \\ PE_{CH_4,SWDS,y} \\ LE_{CH_4,SWDS,y} \end{array} \right\} = \varphi_y \times (1 - f_y) \times GWP_{CH_4} \times (1 - OX) \times \frac{16}{12} \times F \times DOC_{f,y} \\
 \times MCF_y \times \sum_{x=1}^y \sum_j (W_{j,x} \times DOC_j \times e^{-k_j \times (y-x)} \times (1 - e^{-k_j}))$$

(Equation-3)

where,

$BE_{CH_4,SWDS,y}$ = Baseline, project or leakage methane emissions occurring in year generated from waste disposal at a SWDS during a time period ending in year y (tCO₂e/yr)

x = Years in the time period in which waste is disposed at the SWDS, from the first year in the time period (x = 1) to year y (x = y)

y = Year of the crediting period for which methane emissions are calculated (y is a consecutive period of 12 months)

$DOC_{f,y}$ = Fraction of degradable organic carbon (DOC) that can decomposes under the specific conditions occurring in the SWDS for year y (weight fraction)

$W_{j,x}$ = Amount of solid waste type j disposed or prevented from disposal in the SWDS in the year x (t)

ϕ_y = Model correction factor to account for model uncertainties for year y

f_y = Fraction of methane captured at the SWDS and flared, combusted or used in another manner that prevents the emissions of methane to the atmosphere in year y

GWP_{CH_4} = Global Warming Potential of methane

OX = Oxidation factor (reflecting the amount of methane from SWDS that is oxidized in the soil or other material covering the waste)

F = Fraction of methane in the SWDS gas (volume fraction)

MCF_y = Methane correction factor

DOC_j = Fraction of degradable organic carbon in the waste type j (weight fraction)

k_j = Decay rate for the waste type j (1 / yr)

j = Type of residual waste or types of waste in the MSW

Ex-post determination of $F_{CH_4,PJ,y}$

There is no such any legal requirement to destroy the methane as per the laws in Turkey²³. Also as PP monitoring the amount of methane recovered from the landfill site, thus $F_{CH_4,PJ,y}$ is calculated below mentioned formula: -

As per the para 20 equation 5 of methodology AMS-III-G Version 10.0. :-

$F_{CH_4,PJ,y}$ = Methane captured and destroyed/gainfully used by the project activity in the year y (tCH₄)

$$F_{CH_4,PJ,y} = D_{CH_4,y} \times w_{CH_4,y} \times \sum_i LFG_{i,y}$$

Where:

²³ [https://uk.practicallaw.thomsonreuters.com/7-522-2040?contextData=\(sc.Default\)&transitionType=Default&firstPage=true#co_anchor_a539478](https://uk.practicallaw.thomsonreuters.com/7-522-2040?contextData=(sc.Default)&transitionType=Default&firstPage=true#co_anchor_a539478)

2040?contextData=(sc.Default)&transitionType=Default&firstPage=true#co_anchor_a539478

- $D_{CH_4,y}$ = Density of methane at the temperature and pressure of the landfill gas in year y (tons/m³). If $LFG_{i,y}$ is reported at normal conditions of temperature and pressure, the density of methane is also determined at normal conditions
- $w_{CH_4,y}$ = Methane content in landfill gas in year y (volume fraction, m³CH₄/m³LFG). Landfill gas composition shall be measured either on a dry basis or at the same humidity as used to determine $LFG_{i,y}$
- $LFG_{i,y}$ = Landfill gas destroyed via method i (flaring, fueling, combustion, injection to a grid, etc.) in year y (m³LFG). The flow or volume measurement shall be made either on a dry basis or at the same humidity as $w_{CH_4,y}$

a. Baseline Emissions from Electricity Generated (BE_y)

As per AMS-I.D (version 18.0), Baseline emissions associated with electricity generation (BE_y) is calculated using equation (4) below.

$$BE_y = EG_{PJ,y} * EF_{grid,y} \quad (\text{Equation 4})$$

Where:

- BE_y = Baseline emissions in year y (tCO₂e)
- EG_{PJ,y} = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh)
- EF_{grid,y} = Combined margin CO₂ emission factor for grid connected power generation in year y calculated using the latest version of the “Tool to calculate the emission factor for an electricity system” (tCO₂/MWh)

Parameter	Description	Unit	Value	
BE _{EG,y}	Baseline emissions associated with electricity generation in year y	tCO ₂ e/yr	2021	2,952
			2022	5,888
			2023	5,887
			2024	11,775
			2025	11,775
			2026	11,775
			2027	11,775
			2028	5,871
			Average	9,671
BE _{CH₄,SWDS,y}	Baseline methane emissions occurring in year y generated from waste disposal at a SWDS during a time period ending in year y	tCO ₂ e/yr	2021	23,531
			2022	28,067
			2023	32,117
			2024	35,763
			2025	38,992
			2026	41,870
			2027	44,455
			2028	46,793
			2021	23,531
			2022	28,067

$EG_{BL,y}$	Quantity of net electricity supplied to the grid from project activity in year y – Phase 1 (2021-2024)	MWh/yr	10,605
$EG_{BL,y}$	Quantity of net electricity supplied to the grid from project activity in year y – Phase II (2024-2028)	MWh/yr	21,210
$EF_{CO_2,grid,y}$	Emission factor of the grid (tCO _{2e} /MWh)	tCO _{2e} /MWh	0.5552

Baseline emissions from electricity displacement ($BE_{electricity,y}$)

This section demonstrates the calculation of baseline emissions associated with electricity generation using equation (1) of the applied AMS-I.D. methodology.

$$BE_{y,electricity} = EG_{PJ,y} + EF_{grid,y} \quad \text{(Equation-4)}$$

Where;

$$\begin{aligned}
 BE_{electricity,y} &= \text{Baseline emissions associated with electricity displacement } y \text{ (tCO}_{2e}/y) \\
 EG_{PJ,y} &= \text{Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year } y \text{ (MWh)} \\
 EF_{grid,y} &= \text{Combined margin CO}_2 \text{ emission factor of grid connected power generation in year } y \text{ calculated using the latest version of the "Tool to calculate the emission factor for an electricity system" (tCO}_2/\text{MWh)}
 \end{aligned}$$

Calculation of the emission factor

Following the “Tool to calculate the emission factor for an electricity system, version 07.0”, the baseline grid emission factor has been updated as per “Turkey National Grid Emission Factor Information Form”.

The updated operating margin (OM) emission factor in the Joint PD and MR is 0.7424 tCO₂/MWh and the build margin (BM) emission factor is 0.3680 tCO₂/MWh. The assessment team has confirmed that this is the most recent data publicly available at the time of the validation of the crediting period.

According to the “Tool to calculate the emission factor for an electricity system, version 07.0”, the default weights: $W_{OM}=0.50$ for Operating Margin and $W_{BM}=0.50$ for Build Margin in the crediting period of power generation projects are adopted.

The combined margin emissions factor is calculated as follows:

$$EF_{grid,CM,y} = EF_{grid,OM,y} \times W_{OM} + EF_{grid,BM,y} \times W_{BM}$$

Where:

$EF_{grid,CM,y}$	=	Combined margin CO ₂ emission factor of national grid in year y (tCO ₂ /MWh)
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$EF_{grid,OM,y}$	=	Operating margin CO ₂ emission factor of national grid in year y (tCO ₂ /MWh)
W_{OM}	=	Weighting of operating margin emissions factor (%)
$EF_{grid,BM,y}$	=	Build margin CO ₂ emission factor of national grid in year y (tCO ₂ /MWh)
W_{BM}	=	Weighting of build margin emissions factor (%)

$$EF_{grid,CM,y} = 0.7424 * 0.50 + 0.3680 * 0.50$$

$$= 0.5552 \text{ (tCO}_2\text{/MWh)}$$

The calculation of the $EF_{grid,CM,y}$ has been checked by the assessment team and was found consistent with the latest emission factor calculation Tool /4/.

Therefore,

Parameter	Description	Unit	Value	
$BE_{EG,y}$	Baseline emissions associated with electricity generation in year y	tCO ₂ e/yr	2021	2,952
			2022	5,888
			2023	5,887
			2024	11,775
			2025	11,775
			2026	11,775
			2027	11,775
			2028	5,871
			Average	9,671
$EG_{BL,y}$	Quantity of net electricity supplied to the grid from project activity in year y	MWh/yr	10,605	
$EG_{BL,y}$	Quantity of net electricity supplied to the grid from project activity in year y	MWh/yr	21,210	
$EF_{CO2,grid,y}$	Emission factor of the grid (tCO ₂ e /MWh)	tCO ₂ e /MWh	0.5552	

Quantification of Project emissions (PE_y)

Project emissions includes two parts, which are calculated as per AMS-III.G and AMS-I.D separately:

$$PE_y = PE_{Power,y} + PE_{flare,y} + PE_{process,y}$$

Where:

- PE_y = Project emissions in year y (t CO₂/yr)
- $PE_{Power,y}$ = Emissions from the use of fossil fuel or electricity for the operation of the installed facilities in the year y (t CO₂e)
- $PE_{flare,y}$ = Emissions from flaring or combustion of the landfill gas stream in the year y (t CO₂e)

$PE_{process,y}$ = Emissions from the landfill gas upgrading process in the year y (t CO₂e), determined by following the relevant procedures described in annex 1 of AMS-III.H.

No fossil fuels are used for the operation of the installed facilities, only electricity is used. An storage system is used in case the engines are out of operation and exigencies. The landfill gas upgrading process is not applicable to this project, so it is excluded.

Therefore,

$PE_{process,y} = 0$ and,

Since the project activity does not include upgrading of landfill gas therefore project emission is estimated as

$$PE_y = PE_{power,y} + PE_{flare,y}$$

Determination of project emissions from flare ($PE_{flare,y}$)

The landfill gas is combusted in biogas genset. Project emissions related to flaring will be calculated as per Tool06 "Project emissions from flaring".

There is no separate flaring unit for the project, and all landfill gas is combusted in biogas genset used for energy. Also project activity has gas holder of 3500 m³ capacity to hold gas in case of any emergency.

Thus, $PE_{flare,y} = 0$

Determination of project emissions from landfill gas upgrading ($PE_{process,y}$)

In the project activity there is no landfill gas upgrading and no emission sourced from this. Therefore, $PE_{process,y}$ equals to 0.

Project Emissions from renewable electricity production ($PE_{electricity,y}$) as per AMS-I.D.

As per the paragraph 39 of AMS-I.D., version 18.0., for most renewable energy project activities, $PE_{electricity,y} = 0$. However, for the following categories of project activities, project emissions have to be considered following the procedure described in ACM0002 v21.0:

- (a) Emissions related to the operation of geothermal power plants (e.g., non condensable gases, electricity/fossil fuel consumption);
- (b) Emissions from water reservoirs of hydro power plants.

As per the paragraph 40 of AMS-I.D., CO₂ emissions from on-site consumption of fossil fuels due to the project activity shall be calculated using the latest version of the "Tool to calculate project or leakage CO₂ emissions from fossil fuel combustion". Also, the paragraph 41 of AMS-I.D. states

that in case biomass is sourced from dedicated plantations, the procedures in the tool “Project emissions from cultivation of biomass” shall be used

Since none of the exception stated in the relevant paragraphs of AMS-I.D. mentioned above is applicable to the proposed project activity, $PE_{\text{electricity},y}$ as per AMS-I.D., version 18.0., shall be accounted as 0.

Determination of Leakage emissions:

According to paragraph 20 of AMS-III.G leakage must be considered if the methane recovery technology is transferred from another activity. As this project does not involve transfer of any equipment from another activity therefore the leakage has been considered zero.

Following paragraph 42 of AMS-I.D general guidance on leakage in biomass project activities shall be followed to quantify leakages pertaining to the use of biomass residues. Biomass residues are defined as per CDM glossary of Terms version 09.1 as “non-fossilized and biodegradable organic material originating from plants, animals and micro-organisms which is a by-product, residue or waste stream from agriculture, forestry and related industries”. This project does not pertain to the use of biomass residues. Therefore, no leakage effects need to be accounted for under the AMS-I.D and Leakage emission from the project activity is 0.

EMISSION REDUCTIONS

Estimated Emission reductions are calculated as follows:

$$ER_{y, \text{estimated}} = BE_y - PE_y - LE_y$$

Where:

$ER_{y, \text{estimated}}$ = Estimated emission reduction from both methane destruction and grid displacement (tCO₂)

BE_y = Baseline emissions from both methane destruction and grid displacement (tCO₂)

PE_y = Project emissions from both methane destruction and grid displacement (tCO₂)

LE_y = Leakage emissions from both methane destruction and grid displacement (tCO₂)

As per AMS-III.G v10, paragraph 20, The actual emission reduction achieved by the project activity during the crediting period will be calculated using the amount of methane recovered and destroyed/gainfully used by the project activity, calculated as:

$$ER_{y, \text{calculated}} = (1 - OX) \times (F_{CH_4, PJ, y} - F_{CH_4, BL, y}) \times GWP_{CH_4} - PE_y - LE_y \quad \text{(Equation-10)}$$

Where:

$F_{CH_4, PJ, y}$ = Methane captured and destroyed/gainfully used by the project activity in the year y (tCH₄)

$$F_{CH_4, PJ, y} = D_{CH_4, y} \times w_{CH_4, y} \times \sum_i LFG_{i, y} \quad \text{(Equation-11)}$$

Where:

- LFG_{i,y}** = Landfill gas destroyed via method i (flaring, fuelling, combustion, injection to a grid, etc.) in year y (m³LFG). The flow or volume measurement shall be made either on a dry basis or at the same humidity as w_{CH4,y}
- w_{CH4,y}** = Methane content in landfill gas in year y (volume fraction, m³CH₄/m³LFG). Landfill gas composition shall be measured either on a dry basis or at the same humidity as used to determine LFG_{i,y}
- D_{CH4}** = Density of methane at the temperature and pressure of the landfill gas in year y (tonnes/m³). If LFG_{i,y} is reported at normal conditions of temperature and pressure, the density of methane is also determined at normal conditions

For project activities that utilize the recovered methane for power generation, F_{CH4,PJ,y} may be calculated as follows, based on the amount of monitored electricity generation, without monitoring methane flow and concentration:

$$F_{CH4,PJ,y} = \frac{EG_y \times 3600}{NCV_{CH4} \times EE_y} \times D_{CH4} \times GWP_{CH4}$$

(Equation-12)

Where:

- EG_y** = Electricity generation in year y (MWh)
- 3600** = Conversion factor (1 MWh = 3600 MJ)
- NCV_{CH4}** = NCV of methane (MJ/Nm³) use default value: 35.9 MJ/Nm³
- EE_y** = Energy Conversion Efficiency of the project equipment determined from one of the following options:
- Specification provided by the equipment manufacturer specifically for biogas fuel only if the equipment is designed to utilize biogas as fuel. If the specification provides a range of efficiency values, the highest value of the range shall be used for the calculation
 - Default efficiency of 40 per cent

Year	Estimated Baseline Emissions	Estimated Project Emissions	Estimated Leakage Emissions	Estimated Net Reduction
2021	12,979	1,006	0	12,979
2022	29,743	2,386	0	29,743

2023	33,186	2,730	0	30,456
2024	42,173	3,041	0	39,132
2025	44,917	3,316	0	41,601
2026	47,364	3,562	0	43,802
2027	49,562	3,780	0	45,782
2028	25,703	1,984	0	23,719
TOTAL	2,85,627	21,805	0	285,627
Average	40,803	3,115	0	40,803

The assessment team confirms that all data sources and assumptions are appropriate and calculations are correct, applicable to the project activity and will result in a conservative estimate of the emission reductions. In general, the assessment team is able to confirm the following: -

All assumptions and data used by the project participants are listed in the joint PD & MR (version 05 dated 14-Feb-2024) and/or supporting documents, including their references and sources;

- All documentation used by the project participants as the basis for assumptions and source of data is correctly quoted and interpreted in the joint PD & MR (version 07 dated 04-June-2024);

- All values used in joint PD & MR (version 07 dated 04-June-2024) are considered reasonable in the context of the project activity; - The baseline methodology has been applied correctly to calculate project emissions, baseline emissions, and leakage emissions;

- All estimates of the baseline, project and leakage emissions can be replicated using the data and parameter values provided in joint PD & MR (version 07 dated 04-June-2024).

3.4.7 Methodology Deviations

The assessment team confirms that the Joint PD and MR complies with the requirements in the applied methodology AMS-III.G and AMS-I.D. Hence, there is no any methodology deviation applicable for the project activity.

3.4.8 Monitoring Plan

The project monitoring plan is in compliance with the monitoring AMS-III.G Verion.10 and AMS-I.D Version.18. Since the project is energy generation activity through landfill gas, no indicators have been defined regarding project emissions. The monitor-able action plan for the same has been included in section 6.3 of the Joint PD and MR. Validation team confirms that the project participant is able to implement the monitoring plan.

Parameters determined ex-ante: -

- ϕ_y (Default value for the model correction factor to account for model uncertainties): The project selects Option 1: Use a default value to determine ϕ_{default} in the tool “Emissions from solid waste disposal sites” (version 08.0), and the project belongs to the situation of Application A: The VCS project activity mitigates

methane emissions from a specific existing SWDS, thus 0.75 is chosen for ϕ_{default} . The justification is deemed acceptable by the validation team

- **η_{PJ}** (Efficiency of the LFG capture system that will be installed in the project activity): Value used is 85% The collection system which will be installed on the Project site will be able to collect a certain fraction of the total LFG generation. This collection ratio is affected principally by the Project's technical design, therefore, the value suggested in the feasibility study is used.
- **Oxidation factor (OX)**: PP has chosen a default value for Oxidation factor. The same is as per the methodology AMS-III.G. (version 10.0). The justification was accepted by the validation team
- **Fraction of methane in the SWDS gas (F)**: Default value of 0.5, as per the applicable Methodology Tool "Emissions from solid waste disposal site".
- **Methane correction factor (MCF default)**: PP has chosen a default value 1 for the Methane correction factor according to IPCC 2006 guidelines for national greenhouse gas inventories. The same is as per TOOL 04 "Emissions from solid waste disposal sites." The justification was accepted by the validation team
- **Weight fraction (DOC_{FR})**: Default value for the fraction of degradable organic carbon (DOC) in MSW that decomposes in the SWDS. Default value is 0.5 taken from Tool 04-"Emissions from solid waste disposal sites"
- **k_j (Decay rate for waste type)**: As applicable for wet (MAP>1000 mm) tropical conditions (MAT1>20°C) as per the methodological Tool, the decay rate (k_j) has been selected in accordance with the IPCC 2006 Guidelines. The climate where the project is located has been confirmed from data from Turkey meteorological department²⁴ for Sinop Specifically; VVB confirms that the project location falls under wet tropical conditions with MAP> 1000 and MAT> 20°C)
- **GWP_{CH4}** (Global warming potential of Methane) as 28 in accordance with methodological tool, is valid. The values are sourced from the IPCC fourth assessment report- Direct global warming potentials.
- **F** (Fraction of methane in the SWDS gas (volume fraction)):- Fraction of methane captured at the SWDS has been taken as 0.5 for the throughout the crediting period as per IPCC 2006, guidelines to National Greenhouse Gas Inventories. Same has been considered according to Tool04 - "Emissions from solid waste disposal sites. Found correct thus accepted.
- **ρ_{CH4}** (Density of methane at normal conditions)- 0.00067 t CH₄/m³ CH₄, default value taken from Tool 14 "Project and leakage emissions from anaerobic digesters Version 02.0".
- **NCV_{CH4}** (Net calorific value of methane)- PP has chosen a default value for net calorific value of fuel 35.9 MJ/Nm. The same is fixed for throughout the crediting period as per TOOL 04 "Emissions from solid waste disposal sites." Version 8.1. Same was accepted by the validation team.

²⁴ <https://www.mgm.gov.tr/veridegerlendirme/il-ve-ilceler-istatistik.aspx?k=undefined&m=SINOP>

<https://www.mgm.gov.tr/arastirma/buharlasma.aspx?s=2020>

• **DOC_j** (Fraction of degradable organic carbon)- PP has chosen a default value for fraction of degradable organic carbon in the waste type j (weight fraction) according to IPCC 2006 guidelines for national greenhouse gas inventories. Wood and wood products – 43 Pulp, paper and cardboard (other than sludge) – 40 Food, food waste, beverages and tobacco (other than sludge) – 15 Textiles – 24 Garden, park and yard waste – 20 Glass, plastic, metal, other inert – 0 Sewage sludge – 5. The same is as per TOOL 04 “Emissions from solid waste disposal sites.” The justification was accepted by the validation team.

• **WX** (Total amount of waste disposed in a SWDS in the year x)- Feasibility study report (FSR) has been used for the waste amount. Waste compositions study of incoming waste to the landfill has been used for waste composition. Value for throughout the crediting period has fixed as below;

Waste Amount:

Year	Total (tonnes)
2012	18,250
2013	27,375
2014	28,744
2015	30,181
2016	31,690
2017	33,274
2018	36,269
2019	38,083
2020	43,414
2021	56,438
2022	70,548
2023	71,959
2024	73,398
2025	74,132
2026	74,874
2027	75,622
2028	76,379
2029	77,142
2030	77,914
2031	78,693
2032	79,480
2033	80,275
2034	81,077
2035	81,888
2036	82,707
2037	83,534
2038	84,369
2039	85,213
2040	86,065

Waste Composition:

Waste Type	Weight Fraction (%)
Food	58.59
Paper and Cardboard	7.06
Wood	5.90
Textile	3.66
Garden, yard and park waste	0.00
Inert materials (plastic, metal etc.)	24.79
Total	100.00

- $EF_{grid,CM,y}$ (Combined margin CO₂ emission factor for grid connected power generation in year y) : This ex-ante parameter is referred as the combined emission factor CO₂ emission factor for Turkish grid connected power generation. The value is calculated as per Tool 07- "Tool to calculate the emission factor for an electricity system (Version 07.0)" and fixed 0.5552 tCO₂/MWh for throughout the crediting period. Calculation found correct and in line with the Turkish national electricity network, thus accepted.
- TDL_y (Average technical transmission and distribution losses) : The value for this parameters has been fixed as 10.00%, sourced from ANNUAL DEVELOPMENT OF ELECTRICITY GENERATION- CONSUMPTION AND LOSSES IN TURKEY (1993-2020) (<https://webapi.teias.gov.tr/file/9248a331-e529-47ca-8c49-22f8705f5810?download>). Found correct thus accepted.

Parameters determined ex-post: -

- EG_y (Electricity generation in year y) - The amount of electricity generated by the project activity is monitored continuously by SCADA meter on generator project owner. The amount of electricity measurements is monitored continuously and recorded by project staff monthly basis by using PLC. As an estimated value, 10,605 (until 2024) and 21,210 (after 2024) MWe has been used to calculate estimated emission reduction.
- $EG_{PJ,facility,y}$ (Quantity of net electricity supplied by the project activity to the grid in year y) - Quantity of net electricity supplied by the project activity to the grid in year y. The amount of electricity measured by the energy meter and monitored on monthly basis at project plant by keeping it in record as TEIAS data.

Based on the on-site visit and interviewed with staff from Biokütle Enerji Elektrik Üretim A.Ş., The assessment team confirmed that monitoring parameter has been correctly described in the joint PD & MR (version 3.0 dated 14-June-2023) and in compliance with the methodology AMS-III.G Version 10.0, CDM Validation and Verification Standard for project activities version 03.0, and VCS standard version 4.4.

- EE_y (Energy Conversion Efficiency of the project equipment) - Default value in AMS-III.G v 10 has been used which is 40%. This parameter will be monitored as per updated default value during the future monitoring period. Found correct thus accepted.
- $F_{CH4,BL,R,y}$ (Amount of methane in the LFG which is flared due to a requirement in year y) :- This parameter is referred as the quantity of methane emission that would be flared to comply with national or local safety requirement or legal regulations during the monitoring period which will cover under crediting period. Source of value will be considered as per the regulation made by host country i.e., turkey and

frequency of the value will be as per monitoring period. Parameter found in line with the para 41 and 42 of ACM0001 “Flaring or use of Landfill gas” version 19.0” thus accepted.

3.5 Non-Permanence Risk Analysis

No.	Risk that could lead to material errors, omissions or misstatements	Assessment of the risk		Response to the risk in the verification plan and/or sampling plan
		Risk level	Justification	
NA	NA	NA	NA	NA

4 VERIFICATION FINDINGS

4.1 Accuracy of GHG Emission Reduction and Removal Calculations

Means of verification	The verification team assessed whether the data and calculations of GHG emission reductions achieved resulting from the Joint VCS PD & MR. The verification team has checked whether calculations of baseline GHG emissions, project GHG emissions and leakage GHG emissions have been carried out in accordance with the formulae and methods described in the monitoring plan of the Joint VCS PD & MR.		
Findings	No CAR/CL raised on the section.		
Conclusion	Below Parameters are monitored during current monitoring period: -		
	Parameter	EG,y - Electricity generation in year y.	
	Monitoring Procedure	This parameter is measured on monthly basis during current monitoring period, the amount of net electricity generated by project activity. Total electricity generation by project activity is 8,653 MWh measured during monitoring period. The measurement is obtained from PLC (Energy Meter) reading protocols and is crosschecked with .	
			Information
		Equipment	PLC
		Model	Bender – FP200
Rated Voltage		50 V	
Supply Voltage	24 V		

		VVB team checked confirm that the provided records checked onsite and other information are inline with the PLC (Energy Meter) record provided by PP which found consistent.																								
Parameter	EG_{Pj, facility,,y} - Quantity of electricity supplied by the project activity j in year y.																									
Monitoring Procedure	This parameter measured on monthly basis. During current monitoring period amount of electricity exported from the by the project activity is continuously measured by electricity meters owned and operated by the Turkish grid company. The measurement is obtained from Monthly Meter reading protocols (TEIAS Records) and is crosschecked with the electricity sale data (e.g., EPIAS Records). It has been determined that a total of 8,123 MWh of energy has been exported from the grid during this period. <table><tr><td></td><td>Main Meter</td><td>Spare Meter</td></tr><tr><td>Type of meter</td><td>Makel – C520KMT2556</td><td>Makel – C520KMT2556</td></tr><tr><td>Location of meter</td><td>Plant Premise</td><td>Plant Premise</td></tr><tr><td>Accuracy of meter</td><td>0.5 s</td><td>0.5 s</td></tr><tr><td>Serial number of meters</td><td>80276622</td><td>80272124</td></tr><tr><td>Calibration frequency</td><td>10 years</td><td>10 years</td></tr><tr><td>Date of Calibration/</td><td>26-May-2021</td><td>26-May-2021</td></tr><tr><td>Calibration validity</td><td>25-May-2031</td><td>25-May-2031</td></tr></table> VVB team checked confirm that the provided records of calibration certificate checked onsite and meter number, calibration date, accuracy class are inline with the calibration certificate (Energy Meter) record provided by PP which found consistent.			Main Meter	Spare Meter	Type of meter	Makel – C520KMT2556	Makel – C520KMT2556	Location of meter	Plant Premise	Plant Premise	Accuracy of meter	0.5 s	0.5 s	Serial number of meters	80276622	80272124	Calibration frequency	10 years	10 years	Date of Calibration/	26-May-2021	26-May-2021	Calibration validity	25-May-2031	25-May-2031
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Date of Calibration/	26-May-2021	26-May-2021																								
Calibration validity	25-May-2031	25-May-2031																								
Parameter	EE_y - Energy conversion efficiency of the project equipment																									

	Monitoring Procedure	This parameter is referred as the efficiency of gas engines installed at the project site and value for the current monitoring period has been considered as default value i.e., 40%. Default value has been taken from AMS-III.G v 10. VVB checked and confirm from the AMS-III.G version 10.0, found correct thus accepted.
	Parameter	F_{CH4,BL,y} - Methane emissions that would be captured and destroyed to comply with national or local safety requirement or legal regulations in the year y.
	Monitoring Procedure	This parameter is referred as the methane emissions that would be captured and destroyed to comply with national or local safety requirement or legal regulations in the year y considered as default value i.e., 0. Further, Due to no any provision related to flaring due to excess gas generation or gas flaring in any emergency situation on site. Value has been taken as 0 as case 01 is applicable and same has been verified during the onsite visit by lead auditor that there is no any flaring system available onsite, found correct thus accepted.
	Parameter	p_{reg,y} - Fraction of LFG that is required to be flared due to a requirement in year y
	Monitoring Procedure	This parameter is referred as the Fraction of LFG that is required to be flared due to a host country requirement during monitoring period. During current monitoring period, Value has been considered as zero (0) as there is no regulatory requirement in host country i.e., Turkey related to LFG to flare. Same has been verified during the onsite visit by lead auditor that there is no any flaring system available onsite, found correct thus accepted.
	Applus+ Certification assessment team confirms that no any flaring system available onsite for excess gas generation or any emergency situation. Same has been mentioned in monitoring plan. Monitoring plan contains all necessary parameters which have been clearly described in VCS joint PD & MR and that the means of monitoring described in the monitoring plan complies with the requirements of the methodology.	

Emission Reduction Calculations,

$$ER_{y,calculated} = (1 - OX) \times (F_{CH4,PJ,y} - F_{CH4,BL,y}) \times GWP_{CH4} - PE_y - LE_y$$

Where,

$$F_{CH_4,PJ,y} = \frac{EG_y \times 3600}{NCV_{CH_4} \times EE_y} \times D_{CH_4} \times GWP_{CH_4}$$

EG _y	=	Electricity generation in year y (MWh)		
OX	=	Oxidation factor (reflecting the amount of methane from SWDS that is oxidized in the soil or other material covering the waste)	0.1	Dimensionless
F _{CH₄,BL,y}	=	Methane emissions that would be captured and destroyed to comply with national or local safety requirement or legal regulations in the year y	0	tCH ₄
D _{CH₄}	=	Density of methane at the temperature and pressure of the landfill gas in year y	0.00067	ton/m ³
GWP _{CH₄}	=	Global Warming Potential of methane	28	
NCV _{CH₄}	=	Net calorific value of methane	35.9	MJ/Nm ³
EE _y	=	Energy conversion efficiency	40%	%

Project Emission Calculation: -

Project Emissions Related to Electricity or Fossil Fuel Use

PE_{EC,y} = The value is zero since the project generated more energy than it uses and consumed energy has excluded for AMS-I.D.

Project Emissions Related to Flare

PE_{Flame} = As per para 24 of AMS III.G, "...When the amount of methane combusted for energy and the amount that is flared are monitored separately, or when only the landfill gas flow to the flare is monitored and the landfill gas used for energy is calculated based on electricity generation, a destruction efficiency of 100 per cent can be used for the amount that is combusted for energy."

Further, during site visit, VVB team confirmed that there is no separate flaring unit to flare methane in emergency and over production. Since η_{flare} is 100% since emission reduction has been calculated over energy generation and there is no separate flare unit other than biogas genset. Therefore, project emission related to flare is zero (0).

Project Emissions Related to Process Upgrade

PE_{Process}= Since there is no upgrade process for LFG, project emissions related to process is zero

Leakage emission calculation,

Since there is not transfer of equipment therefore leakage emission is considered as 0

Months	Gross Electricity Generations (MWh)	Methane captured and gainfully destroyed ($F_{CH_4,PJ,y}$)	Baseline Emission (BE_{CH_4})	Project Emissions	Leakage Emissions	Net Emission Reduction (ERy)
Jul-21	272.75	45.81	1,154	0	0	1,154
Aug-21	498.79	83.78	2,111	0	0	2,111
Sep-21	522.67	87.79	2,212	0	0	2,212
Oct-21	470.92	79.10	1,993	0	0	1,993
Nov-21	436.93	73.39	1,849	0	0	1,849
Dec-21	420.39	70.61	1,779	0	0	1,779
Sum 2021	2,622.44	440.48	11,100	0	0	11,100
Jan-22	441.82	74.21	1,870	0	0	1,870
Feb-22	422.01	70.88	1,786	0	0	1,786
Mar-22	450.74	75.71	1,908	0	0	1,908
Apr-22	428.12	71.91	1,812	0	0	1,812
May-22	418.30	70.26	1,771	0	0	1,771
Jun-22	446.22	74.95	1,889	0	0	1,889
Jul-22	506.04	85.00	2,142	0	0	2,142
Aug-22	457.29	76.81	1,936	0	0	1,936
Sep-22	415.79	69.84	1,760	0	0	1,760
Oct-22	371.22	62.35	1,571	0	0	1,571
Nov-22	364.69	61.26	1,544	0	0	1,544
Dec-22	435.91	73.22	1,845	0	0	1,845
Sum 2022	5,158.16	866.40	21,833	0	0	21,833
Jan-23	430.34	72.28	1,822	0	0	1,822
Feb-23	441.70	74.19	1,870	0	0	1,870
Sum 2023	872.04	146.47	3,691	0	0	3,691
Total Sum 29-June-2021 to 28-February-2023	8,653	1,453	36,624	0	0	36,624

Net electricity supplied to the grid [MWh]	Baseline emissions or baseline net GHG removals by sinks (tCO ₂ e)	Project emissions or actual net GHG removals by sinks (tCO ₂ e)	Leakage (tCO ₂ e)	Emission reductions or net anthropogenic GHG removals by sinks (tCO ₂ e)
255.30	142	0	0	142
470.37	261	0	0	261
493.60	274	0	0	274
444.01	247	0	0	247
411.84	229	0	0	229
395.74	220	0	0	220
2,470.86	1,371	0	0	1,371

416.09	231	0	0	231
397.84	221	0	0	221
424.57	236	0	0	236
403.15	224	0	0	224
393.41	218	0	0	218
420.56	233	0	0	233
477.74	265	0	0	265
430.64	239	0	0	239
386.00	214	0	0	214
343.71	191	0	0	191
338.45	188	0	0	188
406.36	226	0	0	226
4,838.53	2,686	0	0	2,686
401.25	223	0	0	223
412.32	229	0	0	229
813.57	451	0	0	451
8,123	4,508	0	0	4,508

Months	Net Emission Reduction by electricity (as per AMS-I.D.) (tCO2e)	Net Emission Reduction by LFG capture (as per AMS-III.G.) (tCO2e)	Total Net Emission Reduction (tCO2e)
Jul-21	142	1,154	1,296
Aug-21	261	2,111	2,372
Sep-21	274	2,212	2,486
Oct-21	247	1,993	2,240
Nov-21	229	1,849	2,078
Dec-21	220	1,779	1,999
Sum 2021	1,371	11,100	12,471
Jan-22	231	1,870	2,101
Feb-22	221	1,786	2,007
Mar-22	236	1,908	2,144
Apr-22	224	1,812	2,036
May-22	218	1,771	1,989
Jun-22	233	1,889	2,122
Jul-22	265	2,142	2,407
Aug-22	239	1,936	2,175
Sep-22	214	1,760	1,974
Oct-22	191	1,571	1,762
Nov-22	188	1,544	1,732
Dec-22	226	1,845	2,071
Sum 2022	2,686	21,833	24,519
Jan-23	223	1,822	2,044
Feb-23	229	1,870	2,099

Sum 2023	451	3,691	4,142
Total Sum 01-July-2021 to 28-February-2023	4,508	36,624	41,132

Applus+ Certification is able to confirm that the actual emission reductions reported in this monitoring period are reasonable and appropriate. Applus+ Certification verified the input data for calculating emission reductions and the calculating process, and confirmed the result were complete and transparent. And Applus+ Certification is able to confirm that GHG emission reductions and removals have been quantified correctly in accordance with the monitoring plan and applied methodology for this verification period

4.2 Quality of Evidence to Determine GHG Emission Reductions and Removals

As a result of verification of the ER calculation process, the assessment team confirmed that all the parameters required for the determination of the emission reductions have been included in the joint PD & MR and ER Calculation Spreadsheet of this monitoring period and are consistent with the applied methodology AMS-III.G Version.10 and AMS-I.D Version 18.0 and the monitoring plan contained in the joint PD & MR. The parameters are complete in this monitoring period. After verifying the reported figures with the data sources, it's confirmed that the values of the parameters from the data sources are consistent with those quoted in the ER Calculation Spreadsheet of this monitoring period and joint PD & MR. the verification process for the same has been clearly described above in section 4 of the report. All necessary documentations are collected, referenced and aggregated, which is easily accessible in hard-copy or electronic format. Measurements are performed by calibrated equipment, and the key data can also be cross-checked via other sources, such as records, receipts and inventory data. No assumptions are used that have any material influence on reported emission reductions. LGAI Technological Center S.A. (Applus+ Certification) concludes that during this monitoring period, the evidences for determination of emission reductions are sufficient and reasonable, and the calculation of emission.

Means of verification	The verification team checked monitoring equipment's for the monitoring period. During the verification site visit, the location Energy meter and Gas flow analyzer of the project is also checked. The Calibration details of the meters and analyzer are also checked with calibration certificates. Calibration details of all relevant meters, analyzers checked and found to be ok. Details of calibration provided in Annex. 5 of the report.
Conclusion	CL no 04 were raised on the section and closed successfully.
Conclusion	All parameters which need to be monitored for verification such as amount of electricity generated and other parameters sourced from plant records/ log book also cross checked with stores. The meters installed at plant are

	controlled and supervision of state utility officials. Similarly, Site operator is responsible for monitoring of relevant parameters at plant. It is reported that the data will be kept for 2 years following the end of the crediting period or till the last issuance of VERs for the project activity whichever occurs later. The responsibilities and authorities of project management, data handling and recording, measurement methods and QA/QC procedure have been systematically established and formalized and the same was verified during the site visit. During onsite audit with O&M personnel also confirms that the operational and organizational chart as mentioned in Joint VCS PD & MR is as per the site practice and thus assessment team confirms that the details are correct. The actual emission reduction has been observed to be – 8 % lower (Overall project) in comparison with the ex-ante estimations. The monitoring plan is followed at the project site. The Energy meter were calibrated in line with the registered monitoring plan and there was no delay in calibration observed. Thus, assessment team concluded that the evidences are sufficient in quantity, and appropriate for the quality, to determine the GHG reductions and removals.
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5 VALIDATION AND VERIFICATION OPINION

Validation Conclusion:

Applus+ Certification has been engaged by Sinop Landfill Gas to Electricity Power Plant the Validation and verification of the “Grid Connected Landfill Gas to Energy Power Project by “Sinop İnşaat Sanayi ve Ticaret Limited Şirketi.

The management of the project participant/owner is responsible for the preparation of the GHG emissions data and the reported/estimated GHG emissions reductions on the basis set out within the project’s Monitoring Plan in the Joint VCS PD & MR and the applied methodology /3/ AMS-I. D Version 18.0 and AMS-III.G Version 10.0

Our Validation approach was based on the requirements as defined under the Kyoto Protocol, as well as those defined by the CDM Executive Board and VCS board. Our approach is risk-based, drawing on an understanding of the risks associated with estimated GHG emissions data and the controls in place to mitigate these. The validation can confirm that:

- The projects description compliance with, the requirements of Article 12 of the Kyoto Protocol, the CDM Modalities and Procedures as agreed under decision 3/CMP.1, the annexes to this decision, subsequent decisions and guidance made by COP/MOP & CDM Executive Board and other relevant rules, including the Host Country legislation and sustainability criteria along with VCS guideline version 4.3- and standard version 4.4-
- The project’s baseline and additionality are assessed against AMS-I. D Version 18.0 and AMS-III.G Version 10 /3/.0 for first CP.
- The project’s monitoring plan is assessed against AMS-I. D Version 18.0 and AMS-III.G Version 10 /3/.0 for first CP.
- A risk-based approach has been followed to perform this validation activity. The review of the project description and additional documents related to baseline and monitoring methodology /3/; the subsequent background investigation, follow-up interviews with Project Owner have provided LGAI Technological Center S.A. (Applus+ Certification) with sufficient evidence for positive validation opinion as per the requirement of VCS.

The project is expected to generate 40,803 tCO₂e per year during the length of entire 1st crediting period (01-July-2021 to 30-June-2028).

The VVB can conclude that the project is likely to achieve estimated GHG emission reduction or removals based on a positive conclusion of the reasonableness of assumptions, limitations and methods that support the estimations of the future activities, provided that actual results may vary since the estimates are based on assumptions that are subject to potential changes during the implementation and monitoring of the project.

Verification Conclusion:

Our Verification approach was based on the requirements as defined under the Kyoto Protocol, Marrakesh accord, as well as those defined by the CDM Executive Board. Our approach is risk-based, drawing on an understanding of the risks associated with reporting GHG emissions data and the controls in place to mitigate these. The verification can confirm that

- The project is operated as planned and described in the VCS joint PD & MR;
- The monitoring plan is as per the applied methodology /3/;
- The monitoring process in Monitoring Report is as per the VCS joint PD & MR.
- The development and maintenance of records and reporting procedures are in accordance with the monitoring plan;
- The installed equipment being essential for generating emission reduction runs reliably and is calibrated appropriately
- The monitoring system is in place and generates GHG emission reductions data;
- The GHG emission reductions are calculated without material misstatements.
- No limitation observed for the present verification

Verification period: 01-July-2021 to 28-February-2023 (first and last date included) under 1st CP. Verified GHG emission reductions and removals in the above verification period are reported the Joint PD & MR Version 07, dated 04-June-2024 found correct.

Verified GHG emission reductions and removals in current periodic verification:

Net Emission Reduction from 01-July-2021 to 28-February-2023 (Monitoring Period under First Crediting Period)

For validation, use the following table:

Year	Estimated GHG emission reductions or removals (tCO ₂ e)
01-July-2021 – 31-December-2021	12,979
01-January-2022 – 31-December-2022	29,743
01-January-2023 – 31-December-2023	33,186
01-January-2024 – 31-December-2024	42,173
01-January-2025 – 31-December-2025	44,917
01-January-2026 – 31-December-2026	47,364
01-January-2027 – 31-December-2027	49,562
01-January-2028 – 30-June-2028	25,703
Total estimated ERs	285,627
Total number of crediting years	7

Average annual ERs

40,803

Year	Baseline emissions or removals (tCO ₂ e)	Project emissions or removals (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Net GHG emission reductions or removals (tCO ₂ e)
01-July-2021 – 31-December 2021	12,471	0	0	12,471
01-January-2022 – 31-December 2022	24,519	0	0	24,519
01-January-2023 – 28 February 2023	4,142	0	0	4,142
Total	41,132	0	0	41,132

Year	Ex-ante emissions reductions /removals	Achieved emissions reductions /removals	Percent difference	Justification for the difference
01-July-2021 – 31-December 2021	12,979 tCO ₂ e.	12,471 tCO ₂ e	-4%	Ex-post calculations are slightly higher than ex-ante emission reductions and they are quite close each other.
01-January-2022 – 31-December 2022	29,743 tCO ₂ e.	24,519 tCO ₂ e	-21%	Ex-ante calculations are slightly higher than ex-post emission reductions. For ex-ante emissions, there are many assumptions to calculate emission reductions, so the difference is acceptable such project.
01-January-2023 – 28-February 2023	5,364 tCO ₂ e.	4,142 tCO ₂ e	-30%	Ex-ante calculations are slightly higher than ex-post emission reductions. For ex-ante emissions, there are many assumptions to

				calculate emission reductions, so the difference is acceptable such project.
Total	48,086 tCO ₂ e.	41,132 tCO ₂ e	-17%	

APPENDIX 1: DOCUMENTS REVIEWED OR REFERENCED

No.	Author	Title	References to the document	Provider
1.	VCS	VCS webpage for the project, VCS ID 3899.	Verra Search Page	-
2.	Energy and Natural Resource Ministry	Commissioning Certificate	01-July-2021	PP
3.	UNFCCC/CDM	Applied Methodologies - a) AMS-III.G.-“Landfill Methane Recovery”, b) AMS-I.D. - “Grid connected renewable electricity generation”,	version 10.0 version 18.0	CDM
4.	UNFCCC/CDM	Applied Tools – 1)Tool 04-Methodological Tool: Emissions from solid waste disposal sites. 2) Tool 05-Tool to calculate baseline, project and/or leakage emissions from electricity consumption 3) Tool 07: “Tool to calculate the emission factor for an electricity system”. 4) Tool 08: “Tool to determine the mass flow of a greenhouse gas in a gaseous stream”. 5) Tool 32-Methodological Tool: Positive lists of technologies.	Version 8.1 Version 3.0 Version 4.0 Version 7.0 Version 3.0 Version 4.0	UNFCCC/CDM
5.	NA	EIA Positive Certificate	16-June-2021	PP
6.	NA	The operational lifetime of the project activity from the manufacturer = (Technical specifications)	Manufacturer technical specifications	PP
7.	PP	Draft VCS PD &MR Final PD and MR ER Sheet	Version 01, 20-December-2022 Version 07, 04-June-2024 Version 05, 14-February-2024	PP

No.	Author	Title	References to the document	Provider
8.	NA	Calibration Certificates of energy meters	26-May-2021	PP
9.	NA	Declaration regarding no participation in other GHG program for the concerned monitoring period	22-March-2023	PP
10.	NA	Local Stakeholder Consultation	18-October-2022	PP
11.	NA	Power Purchase Agreements (PPAs)	-	PP
12.	NA	Breakdown details for the monitoring period	-	PP
13.	NA	Grievance Register maintained at site	-	PP
14.	NA	Technical Specifications of LFG collection system, gas engines & gas balloon from manufacturer/supplier	-	PP
15.	NA	Feasibility Report of Sinop Landfill Gas to Electricity Generation Facility	11-December-2020	PP
16.	NA	Provisional acceptance letter issued by Energy & Natural Resources Ministry	29-June-2021	PP
17.	NA	Generation licenses issued by Energy Market Regulatory Authority	24-June-2021	PP
18.	NA	Connection Agreement to Distribution system	05-June-2021	PP
19.	NA	Single Line Diagram for the project activity	-	PP
20.	NA	Meter testing & sealing protocols	26-05-2021	PP
21.	NA	Monthly EPIAS web-portal screenshot for electricity export & import covering monitoring period	-	PP
22.	NA	TEIAS monthly electricity generation records covering monitoring period	-	PP
23.	NA	Training records of the staff involved in operation & maintenance of the plant	-	PP
24.	NA	Social insurance records of the staff involved in operation & maintenance of the plant	-	PP
25.	NA	Land Acquisition agreement between PP & Sinop Municipality	11-December-2020	PP
26.	NA	Records of the annual waste sent to Sinop landfill site provided by Sinop Coastal Municipality Authority	-	PP

APPENDIX 2: CORRECTIVE ACTION REQUESTS, CLARIFICATION REQUESTS AND FORWARD ACTION REQUESTS (CAR/CL/FAR)

CL ID	01	Section no.		Date :02-June-2023
Description of CL				
PP requested to submit pre-feasibility study report on the basis of the daily waste collection & its composition considered for the emission reduction calculations. Further, PP requested to justify any deviation during the monitoring period as project activity is operational since June 2021. Kindly clarify & submit evidences.				
Project participant response				Date: 07-June-2023
1- Feasibility study (FS) has been submitted under Folder no 5. Waste composition is not mentioned in the feasibility study and real and latest composition test data is used. On the other hand, real waste data until 2023 has been submitted now (Folder No 16) which is taken from Union of Municipalities by an official letter. In accordance with the data, FS assumptions are in line with real data except 2021 and 2022. The reason behind this situation might be due to Covid-19, and post covid economic situation of the country that causes decrease in consumption.				
Documentation provided by project participant				
1-Latest waste composition test has been added				
VVB assessment				Date: 12-June-2023
PP has submitted the Pre-feasibility report as per requirement of Verification team but clarify that data of waste collection and composition is not mentioned in it. For daily waste collection information, PP submitted the letter of Union of municipalities dated on 08-June-2023 which specify the daily quantity of MSW during 2012 to 2022 according to weighbridge data. Also provided the details of waste composition test carried out on incoming waste to landfill site which is considered for emission reduction calculation. Hence CL#01 is Closed .				

CL ID	02	Section no.		Date :02-June-2023
Description of CL				
PP requested to clarify the commissioning dates of the landfill facility & Gas Engines and thus start date of the project activity inline with the VCS Version 4.4. Kindly clarify.				
Project participant response				Date : 07-June-2023
1- Provisional acceptance protocol has been submitted. As the official provisional acceptance date is 29/06/2021, the date is taken as start date of project activity. Actual generation has started in 7 th of June but cannot be proven since there is no official document that shows it. The gap is not too much so it is better to keep it 29/06/2021.				
Documentation provided by project participant				
-				

VVB assessment	Date: 12-June-2023
PP has provided the provisional acceptance letter dated on 01-July-2021 having installation capacity 1.414 Mwe. PP has mentioned that installation of second generator will be done till end of 2024 in section 1.1 of revised PD & MR which is acceptable. Hence CL#4 is closed.	

CL ID	03	Section no.	3.1	Date:02-June-2023
Description of CAR				
PP is requested to submit the supporting for the SDG targets mentioned in Table: 4 Sustainable Development Contributions under section 1.17.2 of VCS Joint PD & MR to assessment team. Kindly submit.				
Project participant response				Date: 07-June-2023
For SDG 13, waste amount and composition and gross electricity generations provided. For SDG 8, employment records has been provided. For SDG 7, net electricity supplied to grid data has been provided. For SDG 6, avoidance of cooling water use has been provided in ER excel with references.				
Documentation provided by project participant				
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VVB assessment	Date: 12-June-2023
PP has provided, 1. Letter of Union of municipalities dated on 08-June-2023 which specify the daily quantity of MSW during 2012 to 2022 according to weighbridge data used for electricity generation and calculation of emission reduction. 2. Education Participation Form dated on 16-December-2022 submitted to Verification team as supporting document for SDG 8 employee training. 3. EPAIS Screenshots of monthly electricity generation record submitted to verification team as supporting document for SDG 7. 4. No any evidence provided related to SDG 6 Hence CL#03 is open.	
Project participant response	
4- SDG 6 has been removed now.	
VVB assessment	Date: 14-June-2023
PP has provided the revised PD & MR to verification team. During verification assessment team observed that PP has removed the SDG 6 from PD & MR as it is irrelevant with project activity. Hence CL #03 is closed.	

CL ID	04	Section no.	3.4.8	Date:02-June-2023
Description of CAR				
The complete information of metering equipment like serial number, make, accuracy class, calibration date and validity of calibration and calibration frequency is mentioned in Section 7.1 of VCS Joint PD & MR. Further, to verify, calibration dates supporting documents of electricity meters are also not submitted. Kindly submit.				
Project participant response				Date: 07-June-2023

1- Information added to PD and documents submitted now.	
Documentation provided by project participant	
-	
VVB assessment	Date: 12-June-2023
PP has provided the PD & MR, during verification assessment team verify that PP has mention energy meter calibration detail in section 7.1 but supporting document is not provided. Hence CL#04 is open.	
Project participant response	
Calibration of meters were done by the first index protocol. Therefore, first index protocol that was submitted before is also the proof of calibration of meter. Test results of the meter can be seen in 2-3-4 pages of the given PDFs named “Main meter instalment protocol” and “ Backup meter instalment protocol”.	
VVB assessment	Date: 14-June-2023
PP has provided the PD & MR, during verification assessment team verify that PP has mention energy meter calibration detail in section 7.1 and calibration certificate is provided as supporting document. Verification team confirms the meter no, certificate date and its accuracy class and found consistent. Hence CL#04 is Closed.	

CL ID	05	Section no.	3.1	Date :02-June-2023
Description of CL				
PP requested to clarify about cover provided over landfill site to collect LFG and reduce the climate effect on project activity. Also provide the diagram and other evidence of covers provided in landfill site.				
Project participant response				Date : 07-June-2023
The landfill is covered by soil with height of 0.3-0.5 m as per requested in “Solid Waste Facilities Field Management and Operating Guideline” prepared by the Ministry of Environment and Urbanization. The diagrams and rules are given in the guideline as well. The guideline has been provided now in Folder				
Documentation provided by project participant				
-				
VVB assessment				Date: 12-June-2023
PP has clarify about the soil covered provided with height of 0.3-0.5 m as per requested in in “Solid Waste Facilities Field Management and Operating Guideline” prepared by the Ministry of Environment and Urbanization. But during verification assessment team observed that information related to soil cover is not mention in revised PD&MR. Also supporting document is not provided by PP. Hence CL#5 is open.				
Project participant response				
Related information is now given in section 2.3 “Environmental Impact”.				
VVB assessment				Date: 14-June-2023

PP has provided the PD & MR, during verification assessment team verify that PP has mention the height of landfill cover of soil provided over the landfill site in section 2.3 of revised PD & MR and provided the “Solid Waste Facilities Field Management and Operating Guideline” published by the Ministry of Environment and Urbanization. Found consistent with Ministry guideline for landfill cover. Hence **CL#04 is Closed**.

CL ID	06	Section no.	3.3.2	Date :12-June-2023
Description of CL				
PP requested to submit the meeting invitation letter, meeting attendance record and photos of meeting as proof of Local Stakeholder Consultation.				
Project participant response				Date : 07-June-2023
1- Proof documents has been provided.				
Documentation provided by project participant				
1-LSC Photos, Invitation mail, invitation poster, attendance sheet, public comments, logbook received statement.				
VVB assessment				Date: 14-June-2023
PP has provided LSC Consultation photos, Invitation Poster, Invitation email screenshot and attendance record as proof of local stakeholder consultation as per requirement of VCS Joint PD & MR Template guideline 4.2. Which found consistent. Hence CL#05 is Closed .				

CL ID	07	Section no.	3.1	Date :12-June-2023
Description of CL				
Project proponents shall provide evidence to a validating VVB that only the landfill gas recovered in the project is used for power generation; no other gas or fuels except a start-up fuel are used. As per Para 22 AMS III.G				
Project participant response				Date : 07-June-2023
PP statement related to issue has been submitted now.				
Documentation provided by project participant				
-				
VVB assessment				Date: 14-June-2023
PP has provided the declaration letter dated on 14-June-2023 as supporting document for fuel consumption for running landfill to electricity generation plant on Sinop Landfill site as per requirement of para 22 of methodology AMS-III.G. Hence CL#06 is Closed .				

CAR ID	01	Section no.	3.1	Date:02-June-2023
Description of CAR				

1. Date format throughout the VCS Joint PD & MR, dated: 20-December-2022, is not inline with the guidelines to complete VCS Joint PD & MR, Template version 4.2. Thus, corrective action sought. 2. Section 1.1, of monitoring report is found not inline with the VCS MR template v 4.1. Thus, corrective action sought. 3. Various editorial mistakes/insufficient information observed in the VCS Joint PD & MR, dated: 20-December-2022 which are provided in the form of comments. Corrective action sought for the same. 4. The start date of crediting period is found inconsistent in section 1.1 of VCS Joint PD & MR, dated: 20-December-2022. Thus, corrective action sought.	
Project participant response	Date: 07-June-2023
1- Corrected where relevant. 2- Corrected as per Joint PD&MR template v4.2 3- Corrected where relevant. 4- Corrected.	
Documentation provided by project participant	
-	
VVB assessment	Date: 12-June-2023
1. PP has submitted the revised PD & MR to verification team, during assessment verification team observed that date format is not inline with VCS template guideline 4.2 which is acceptable. 2. PP has submitted the revised PD & MR to verification team, during assessment verification team observed that section 1.1 is inline with VCS guideline which is acceptable. 3. Verification team checked the revised PD & MR submitted by the PP. During assessment verification team observed that PP has corrected the editorial mistakes. 4. PP has submitted the revised PD & MR to verification team. During verification assessment team found that start date of crediting period is inline with the VCS PD & MR template guideline 4.2. Hence CAR #01 is open.	
Project participant response	
1-Date format has been correct in entire document.	
VVB assessment	Date: 14-June-2023
PP has provided the revised PD & MR to verification team for assessment. During assessment, team found that date format throughout the PD & MR is changed and provided as per the VCS PD & MR Template guideline 4.2. Which is acceptable. Hence CAR#01 is closed.	

CAR ID	02	Section no.	3.4.6	Date: 02-June-2023
Description of CAR				
No supporting information is found in Appendix X of the VCS Joint PD & MR. PP is requested to Delete this appendix (title and instructions) where no appendix is required. Thus, corrective action sought.				
Project participant response				Date: 07-June-2023
1- Appendix 1 has been added now.				
Documentation provided by project participant				
-				
VVB assessment				Date: 14-June-2023

PP has submitted the revised PD & MR to verification team. During assessment verification team found that PP has mentioned the emission reduction data in appendix. Hence **CAR #2 is closed.**

APPENDIX 3: COMPETENCE OF TEAM MEMBERS AND TECHNICAL REVIEWERS

Verification team member

No.	Role	Type of resource	Last name	First name	Affiliation (e.g. name of central or other office of VVB or outsourced entity)	Involvement in			
						Desk review	On-site inspection	Interview(s)	Verification findings
1.	Lead Auditor/Technical Expert	OR	Takarkhede	Dr. Atul	TQC-Outsourced entity	Yes	Yes	Yes	Yes
2.	Country Expert	OR	Mohan Singh	Mr. Jitendra	TQC-Outsourced entity	Yes	No	N/A	Yes

Technical reviewer and approver of the Joint Validation and verification-certification report

No.	Role	Type of resource	Last name	First name	Affiliation (e.g. name of central or other office of VVB or outsourced entity)
1.	Technical reviewer (TR) / Technical Expert (TE)	EI	Xue	Mr. Denny	Applus+ Certification
2.	Approver	IR	Calle de Miguel	Mr. Agustin	Applus+ Certification

Short CVs of the Team:

- Dr. Atul Takarkhede** Dr. Atul Takarkhede is Ph.D. (Environmental Sciences) from Institute of Science, RTM Nagpur University, Nagpur, and he has already published different technical papers related to environmental sciences. He counts with more than 11 years of experience in field of Environmental Auditing, consulting and accreditation. He is an expert in ISO 9001-14001, CO2/GHG Reporting, Carbon Foot Print, Energy, Water and Waste Management reporting for organizations' environmental performance. His professional portfolio is mainly related with carrying out EIA, conducting QA/QC of EIA Reports; conducting environmental/water audits; NABET requirements appliance, functional area expert in Water Pollution & Solid & Hazardous Waste management among others. Furthermore, he counts with solid experience on CDM-VCS-GS consultancy and auditing. Currently he is associated with True Quality Certifications Private Limited and empanelled with Applus+ Certification to carry out GHG audits in the aforementioned schemes. Dr. Atul Takarkhede is based in Nagpur, India.
- Mr. Jitendra Mohan Singh**, has done Advanced MSc in Sustainable Energy Systems and Management from International Institute of Management, University of Flensburg, Germany and B.Tech. in Agricultural Engineering from Allahabad University, India. He has more than (18) years of working experience in different organizations like IARI, IIT Delhi, ICAR, IRADe, CAPART, SMEC

and Perenia Carbon and M B Power (Madhya Pradesh) Ltd. in the area of Agriculture, Energy & Environment and Climate Change. He also worked on contract basis (ad hoc) as a RIT expert in UNFCCC from 2010 to 2013. Currently, he is empanelled with Applus+ Certification since 2020 and has been involved Verifications of PAs as Lead Auditor and Technical Expert for Renewable and non-Renewable as well as Energy Demand.

- **Mr. Denny Xue**, has a bachelor's degree on Thermal Energy Engineering and Master's Degree on Environmental Engineering. He has more than 10 years of experience on CDM project development. Before he joined Applus+ LGAI, he has been worked for Shanghai Chuanji Investment and Management which is a CDM consultancy company as a project manager for CDM project development. He is working with Applus+ since 2011 carrying out Validation and verification for CDM/GS/VCS project under scope 1 and 13 as auditor, lead auditor, technical expert and technical reviewer.

APPENDIX 4: ABBREVIATIONS

Abbreviations	Full texts
BM	Build Margin
CAR	Corrective Action Request
CDM	Clean Development Mechanism
CER	Certified Emission Reduction(s)
CL	Clarification request
CM	Combined Margin
CMS	Central Monitoring system
CO ₂	Carbon dioxide
CO ₂ e	Carbon dioxide equivalent
DNA	Designated National Authority
DOE	Designated Operational Entity
DR	Document Review
EF	Emission Factor
EIA	Environmental Impact Assessment
ER	Emission Reductions
FAR	Forward Action Request
GHG	Greenhouse gas(es)
GWP	Global Warming potential
PP	Project Participant
VCUs	Verified Carbon Unit